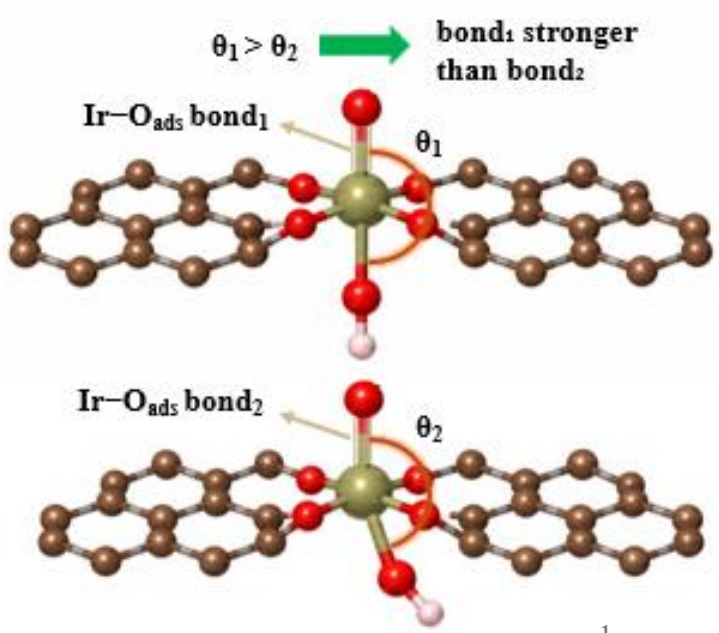
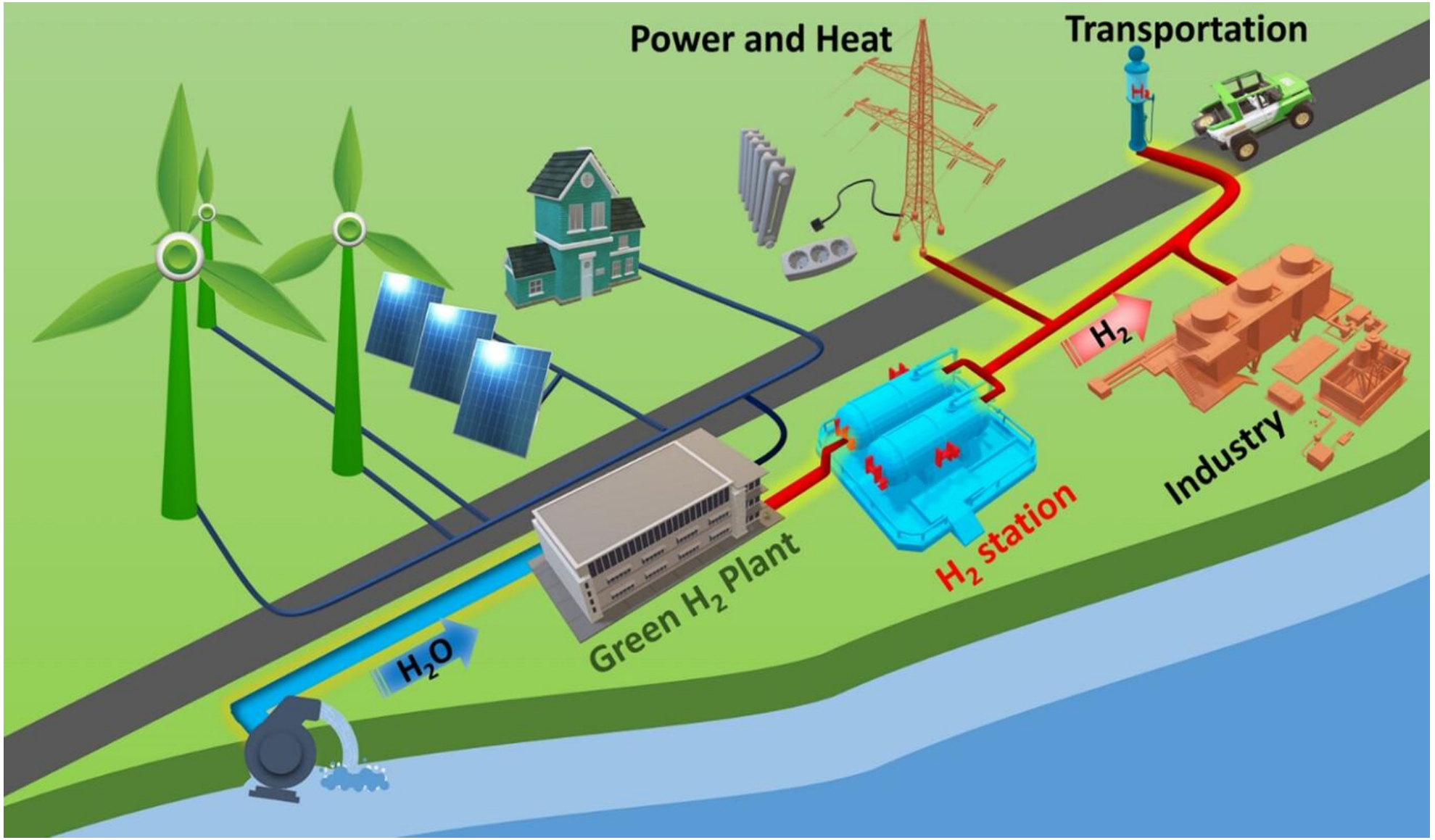
Effect of MO_6 Bond Angle on Metal Oxygen Bonding in Single-Atom Catalysts

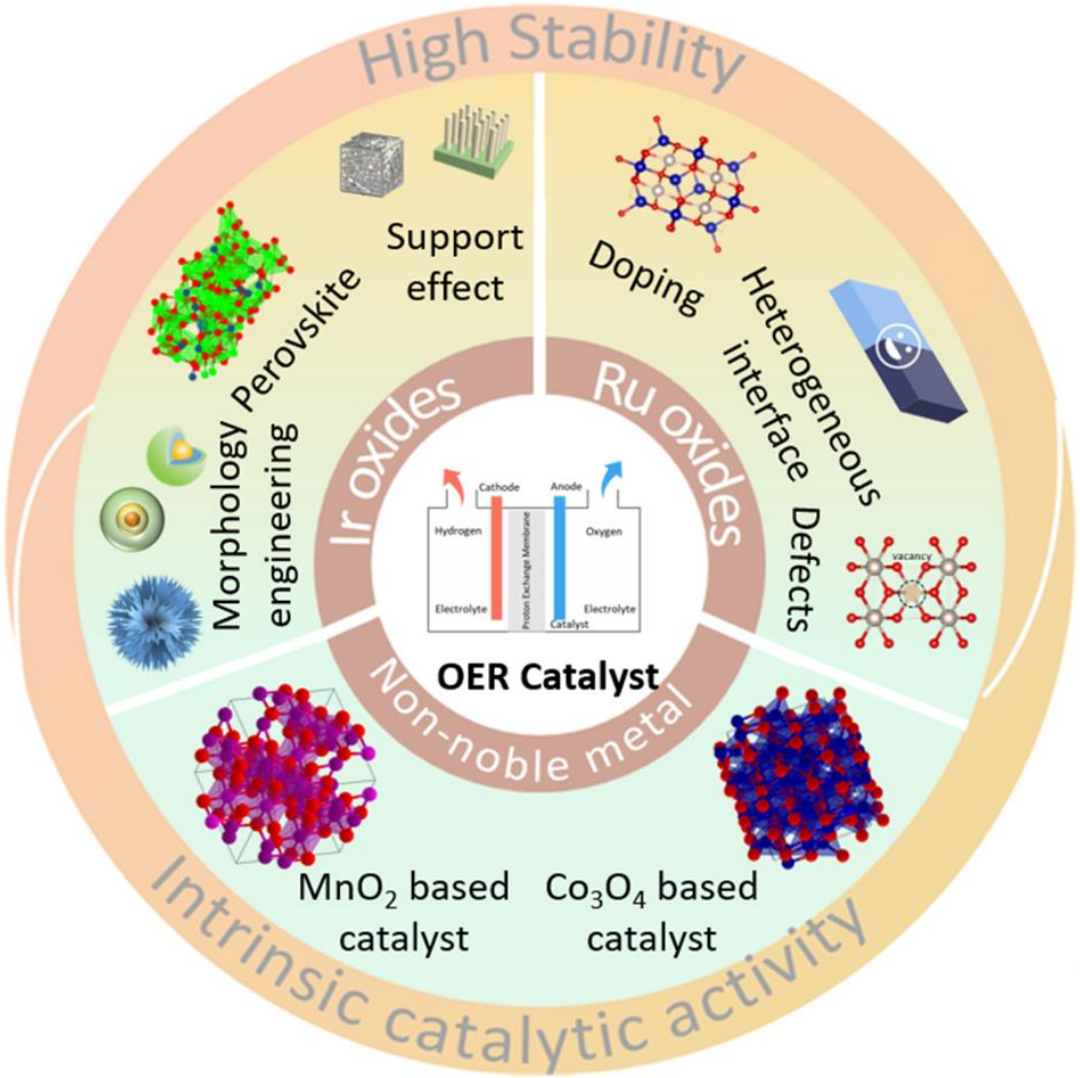


1. Oxygen Evolution Reaction (OER)

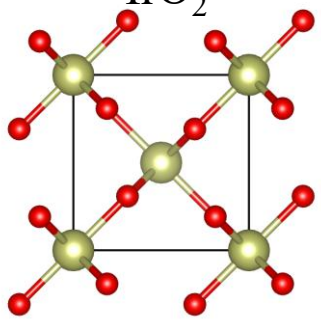


Yu M, Budiyanto E, Tüysüz H. Angew Chem Int Ed Engl. 2022 Jan 3;61(1):e202103824.

2. Widespread Applications of Transition Metal Oxides in OER

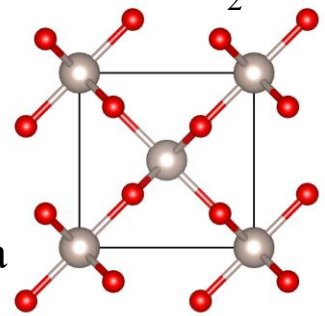


IrO_2



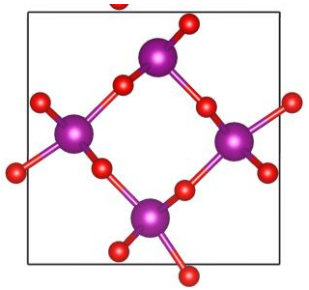
- High catalytic activity
- Good stability in acidic media

RuO_2



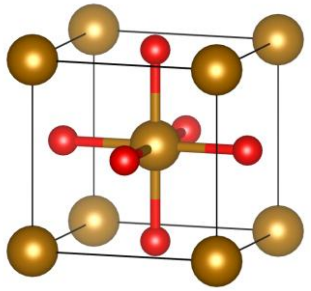
- High catalytic activity
- Prone to dissolution in acidic media

MnO_2



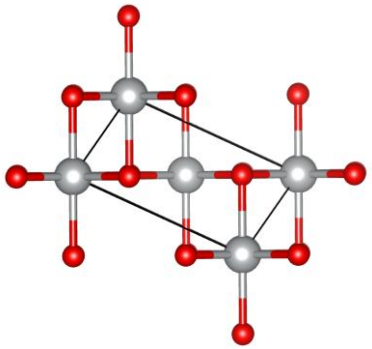
- Active in alkaline media
- Multiple crystal phases

Fe_2O_3



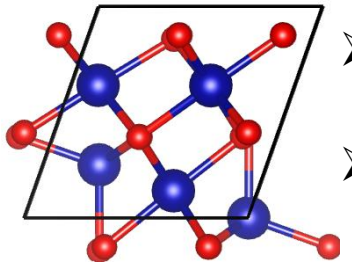
- Active in alkaline media
- Earth-abundant and low-cost

NiO



- Active in alkaline media

Co_3O_4



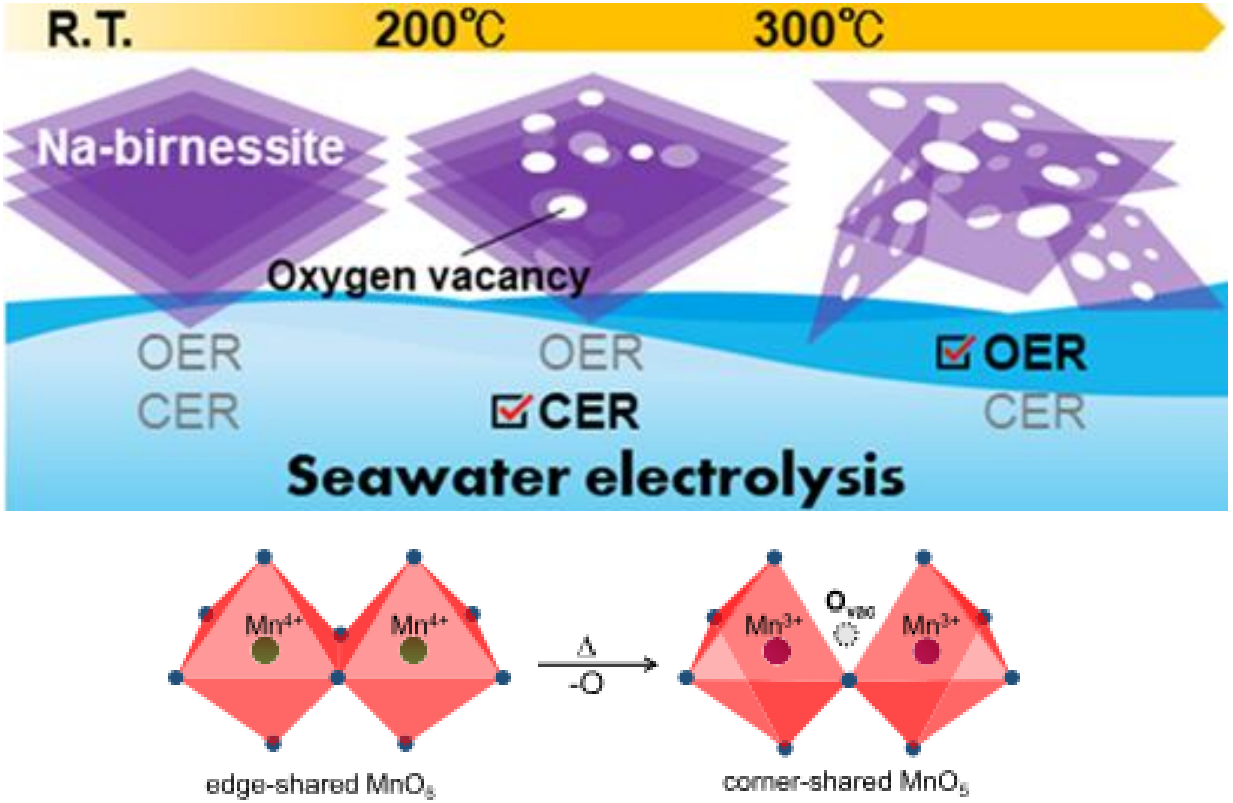
- Active in alkaline media
- Surface reconstruction

Yingying Xu, Yingxia Zhao, et al; Balancing the relationship between the activity and stability of anode oxide-based electrocatalysts in acid for PEMWE electrolyzers. *J. Mater. Chem. A* 2024; 12 (30): 18751–18773.

3. Structural Engineering Strategies to Enhance the OER Activity of Metal Oxides (MO_x)

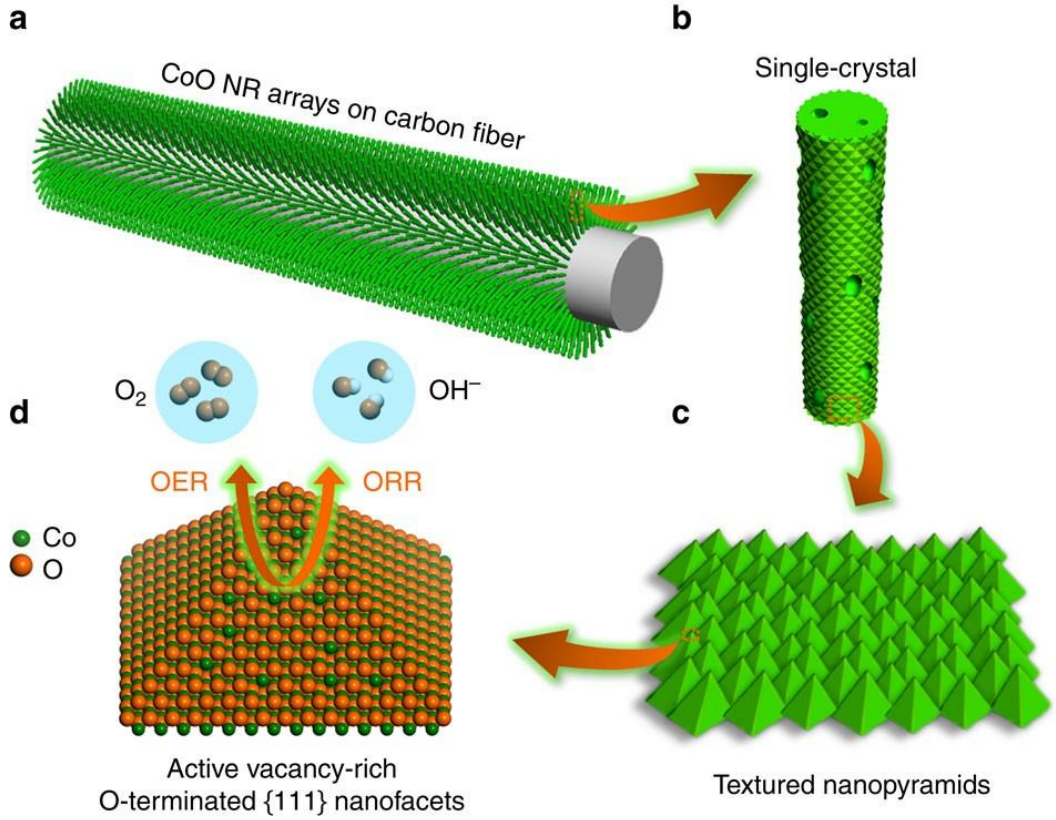
3.1 Introduction of Oxygen Vacancies and Other Defects

MnO_x



Hikaru Abe, Ai Murakami, Shun Tsunekawa, Takuya Okada, Toru Wakabayashi, Masaaki Yoshida, Masaharu Nakayama; Selective Catalyst for Oxygen Evolution in Neutral Brine Electrolysis: An Oxygen-Deficient Manganese Oxide Film. *ACS Catal.* 4 June 2021; 11 (11): 6390–6397.

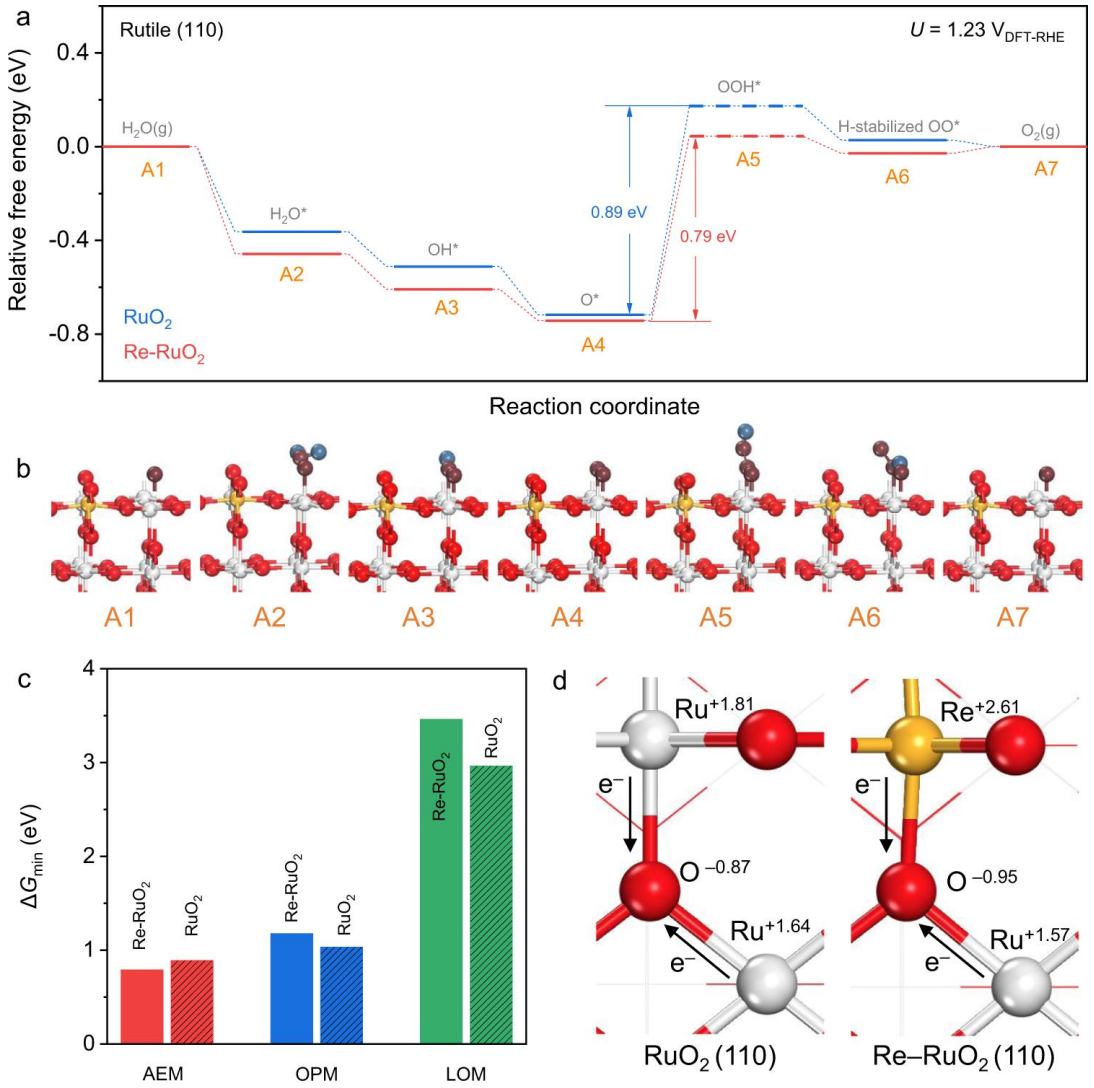
CoO_x



Ling, T., Yan, DY., Jiao, Y. *et al.* Engineering surface atomic structure of single-crystal cobalt (II) oxide nanorods for superior electrocatalysis. *Nat Commun* **7**, 12876 (2016).

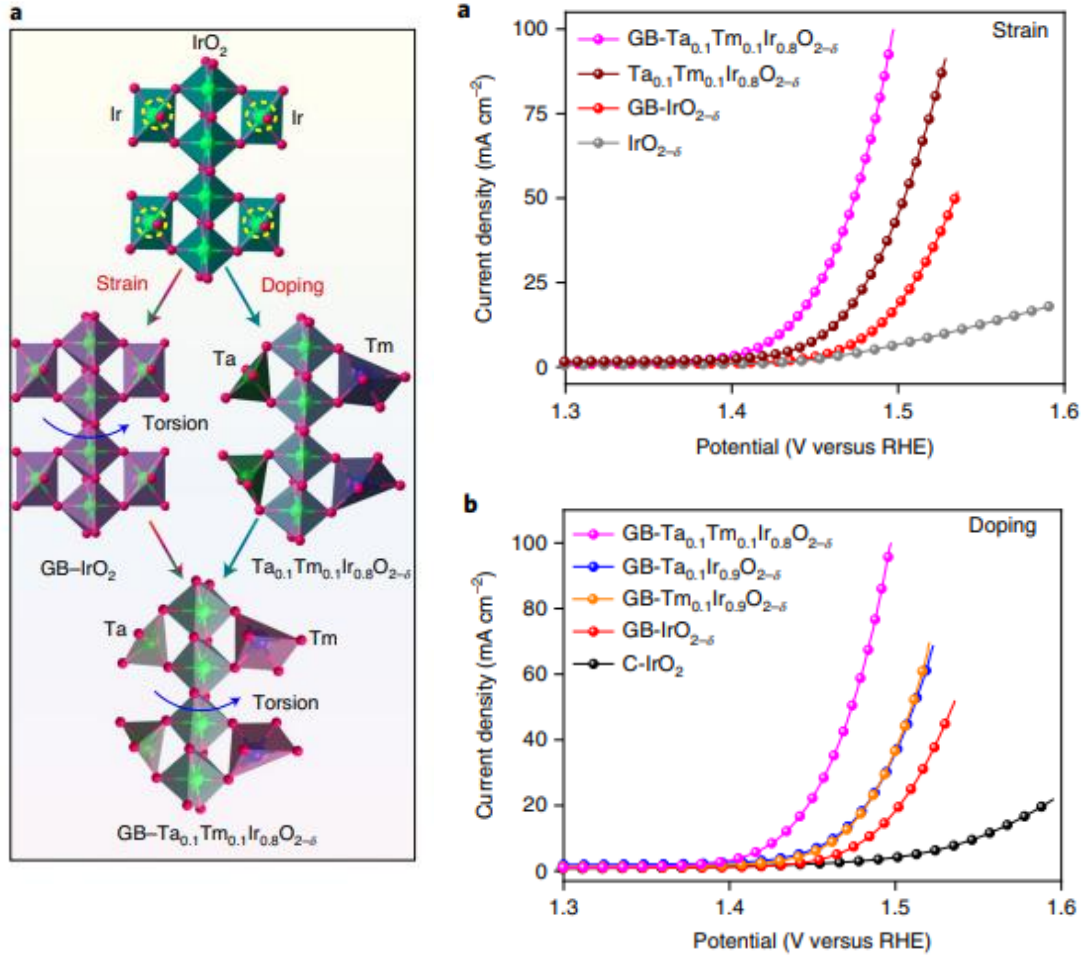
3.2 Elemental Doping

Re doped RuO_2



Jin, H., Liu, X., An, P. *et al.* Dynamic rhenium dopant boosts ruthenium oxide for durable oxygen evolution. *Nat Commun* **14**, 354 (2023). https://doi.org/10.1038/s41467-023-35913-6

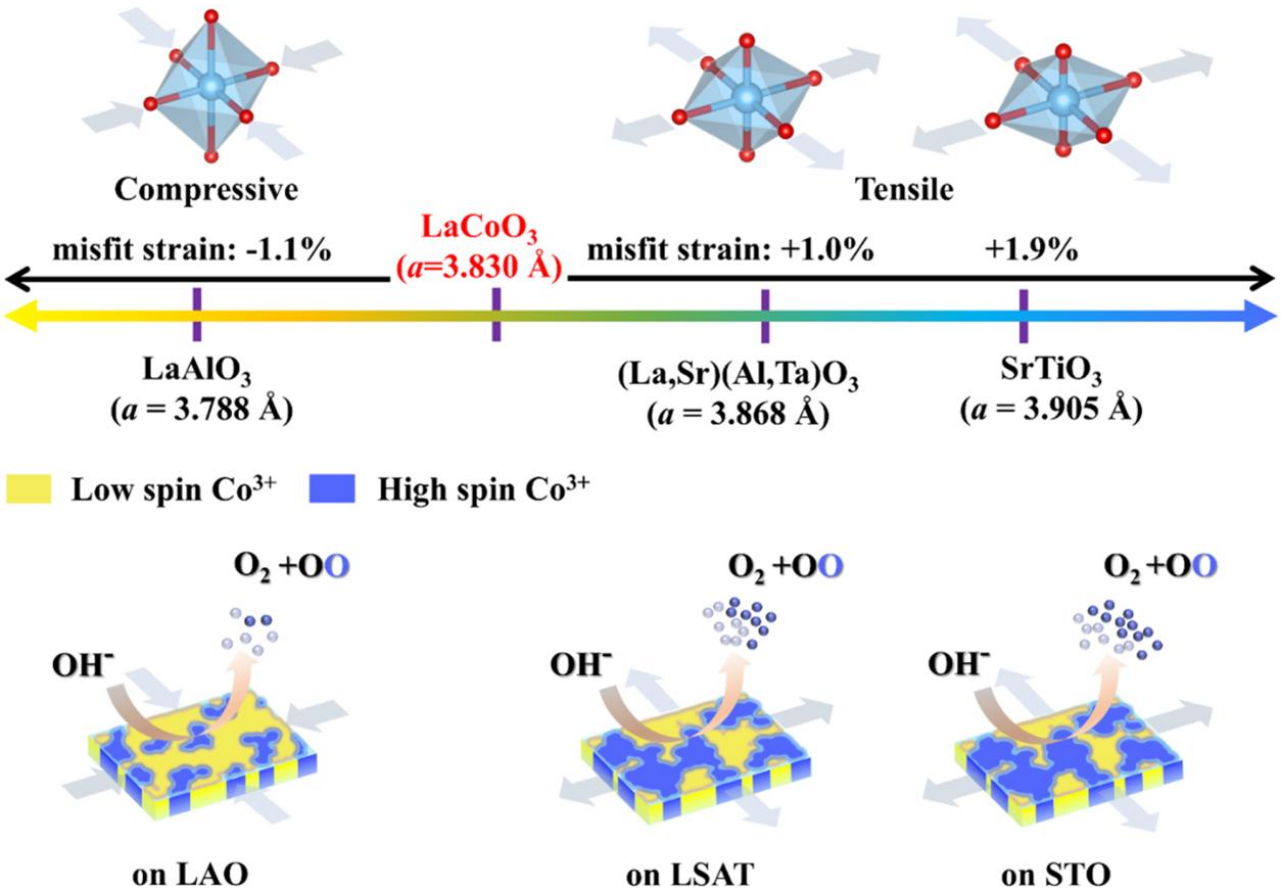
Ta & Tm doped IrO_2



Hao, S., Sheng, H., Liu, M. *et al.* Torsion strained iridium oxide for efficient acidic water oxidation in proton exchange membrane electrolyzers. *Nat. Nanotechnol.* **16**, 1371–1377 (2021). https://doi.org/10.1038/s41565-021-00986-1

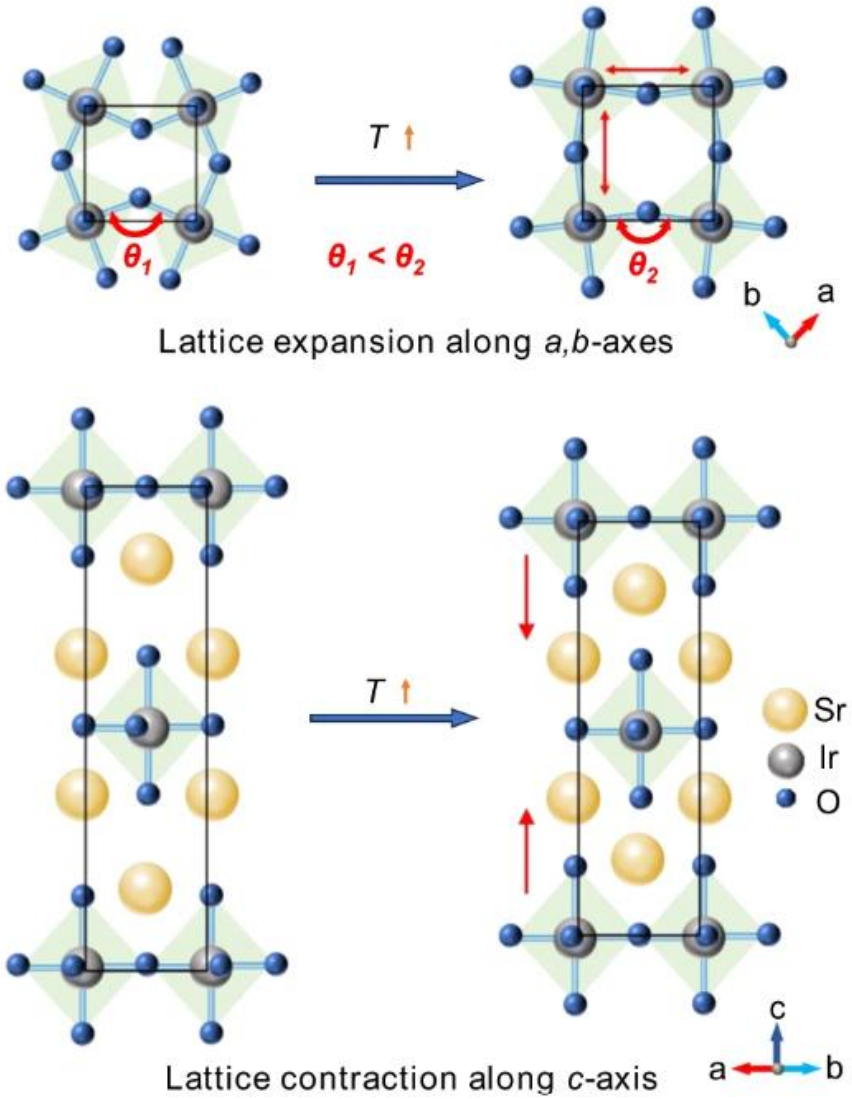
3.3 Strain Engineering

Tensile engineering



Shanquan Chen, Feng-Hui Gong, et al; Unleashing Electrocatalytic Oxygen Evolution Activity: Engineering Spin States in Strained Correlated Oxides for Enhanced Performance. *ACS Nano* 8 July 2025; 19 (26): 23647–23658.

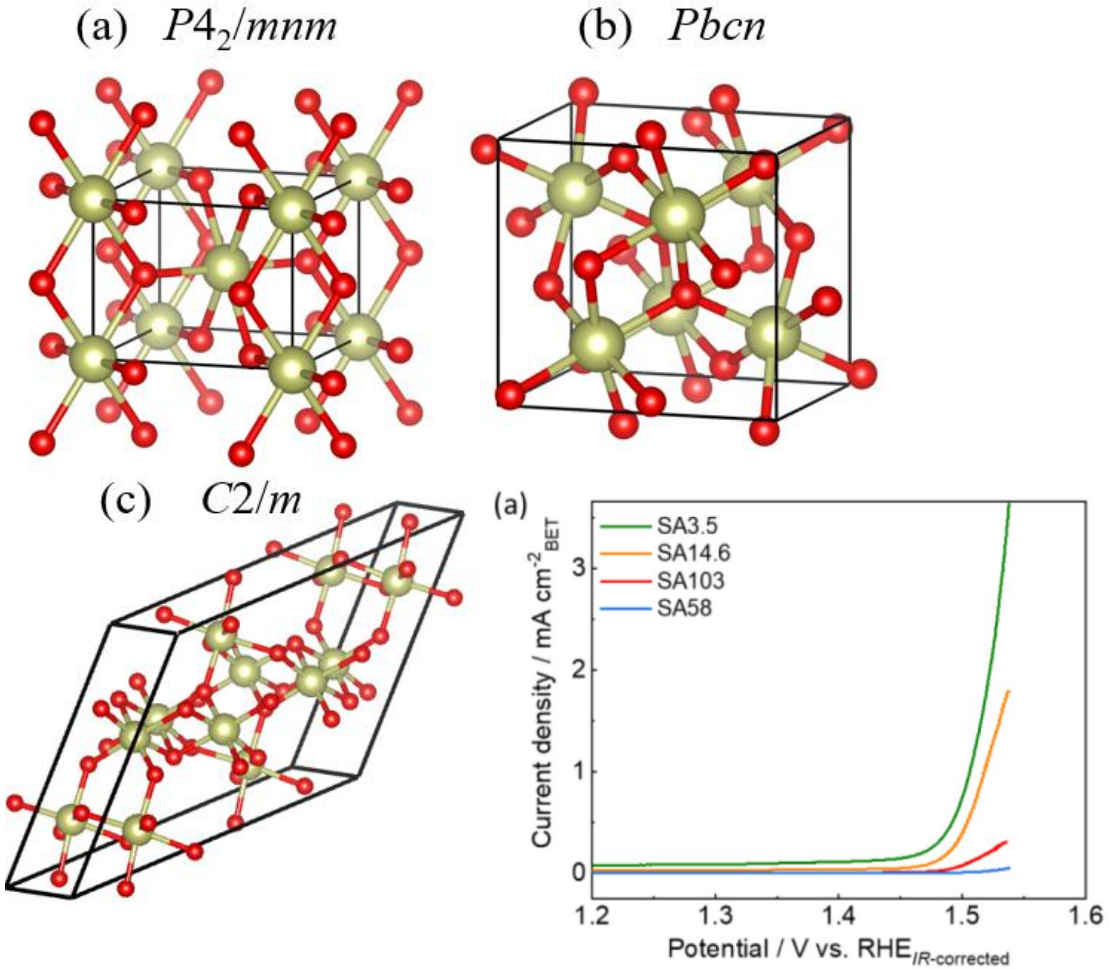
Expansion engineering



Du, Y., Xie, F., Lu, M. *et al.* Continuous strain tuning of oxygen evolution catalysts with anisotropic thermal expansion. *Nat Commun* **15**, 1780 (2024).

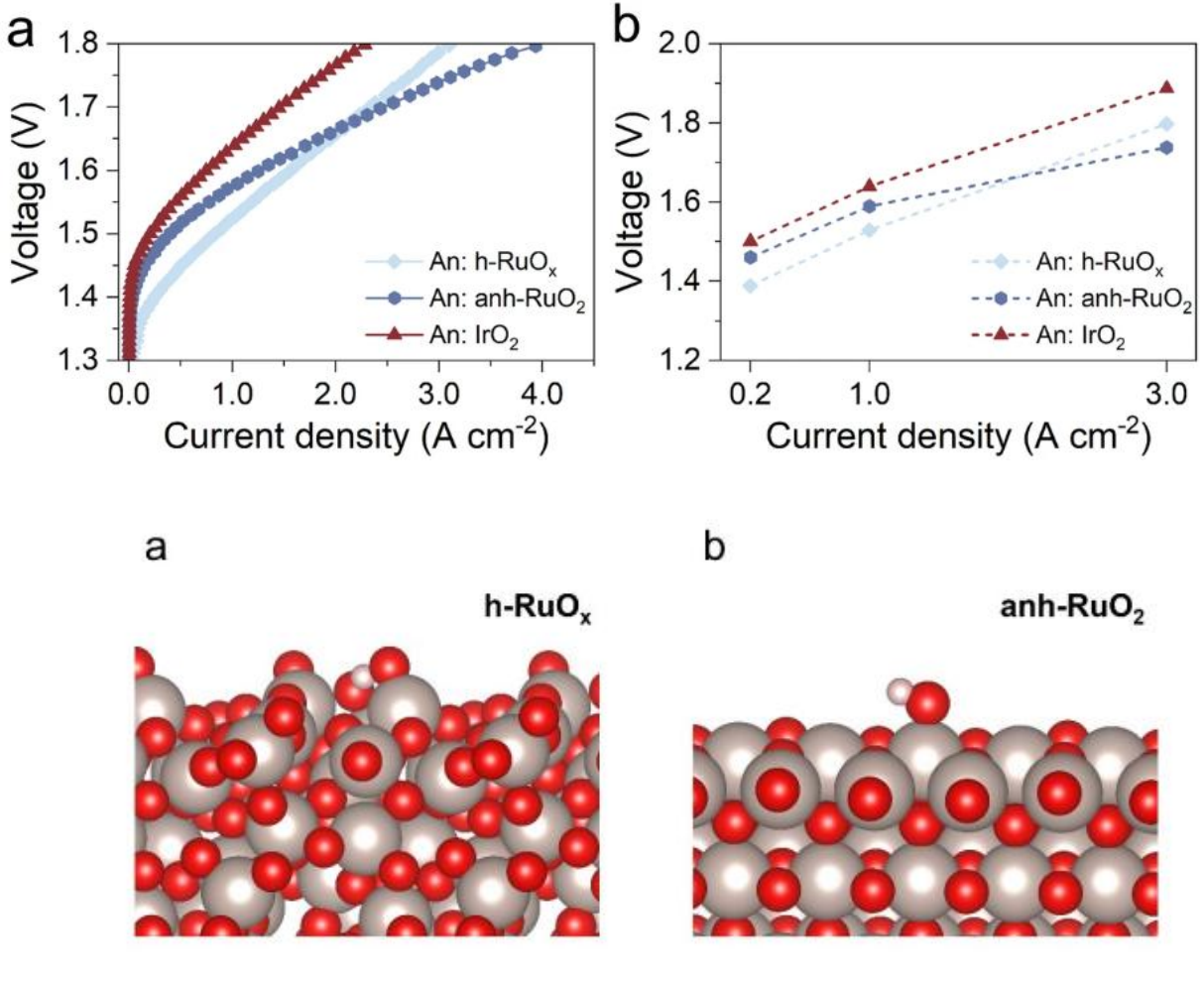
3.4 Crystal Structure Engineering

IrO_2 with 3 space group



Thakur, N.; Ren, Y.; Kumar, M.; et al. Identifying Active Sites of IrO_x Catalysts for OER: A Combined Operando XAS, SEIRAS, and Theoretical Study. J. Am. Chem. Soc. 2025, 147 (34), 30613–30625.

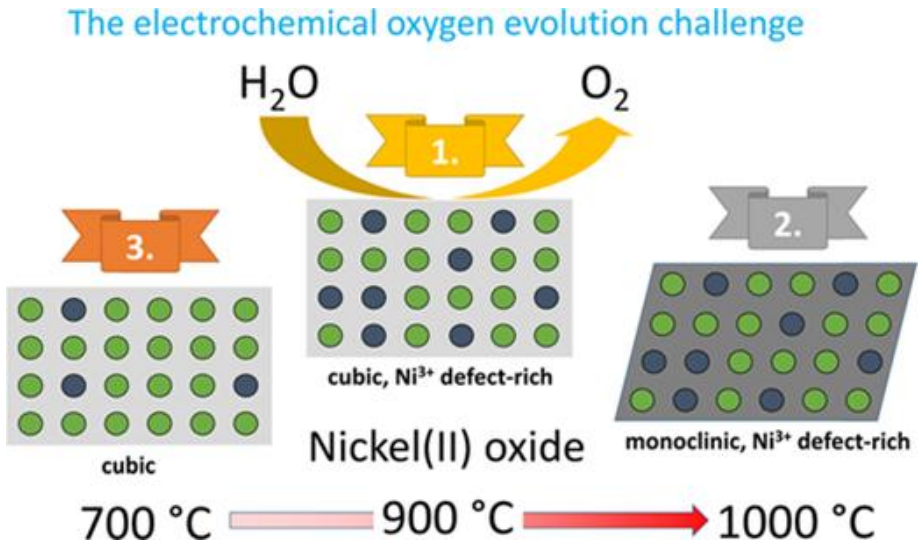
Amorphous phase of RuO_2



Tang, J., Guan, D., Xu, H. *et al.* Undoped ruthenium oxide as a stable catalyst for the acidic oxygen evolution reaction. *Nat Commun* **16**, 801 (2025). https://doi.org/10.1038/s41467-025-56188-z

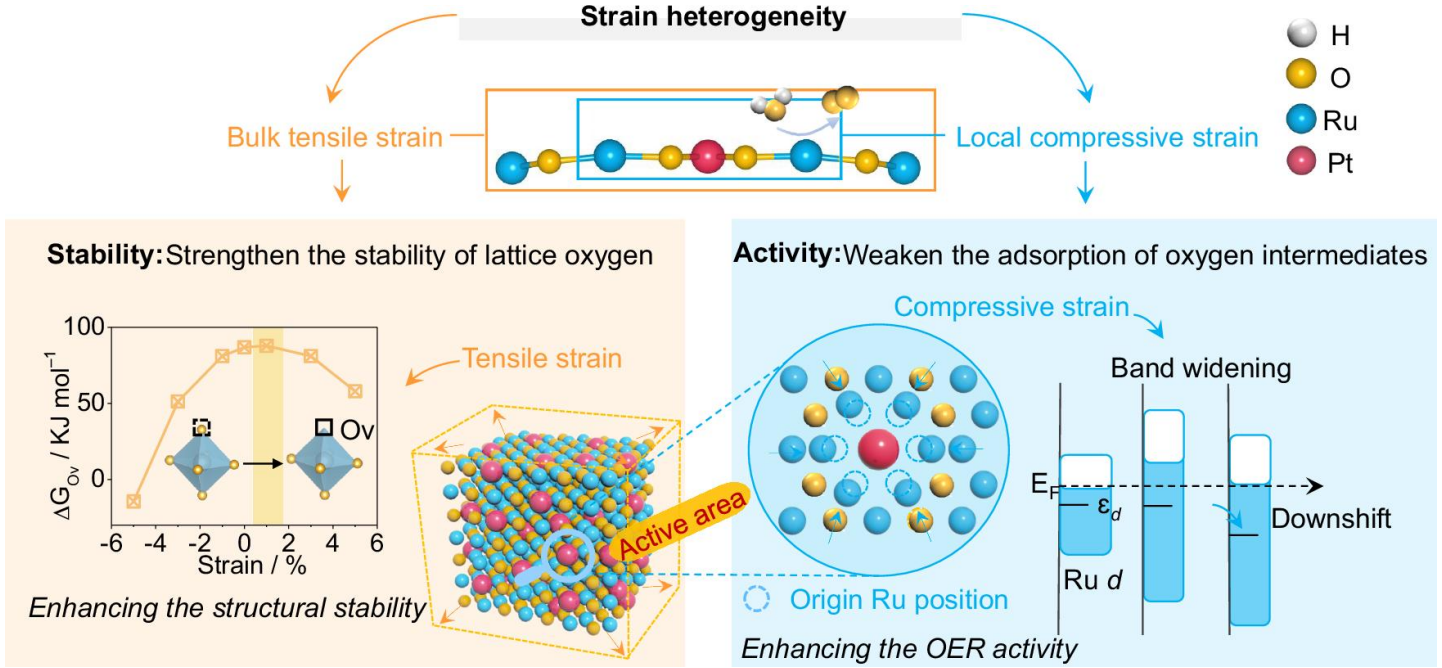
3.5 Combined Structural Engineering

Control defects and space groups in nickel oxides with different temperature to find the structure with the highest OER activity.



Matthias Steimecke, Gerda Seiffarth, Christian Schneemann, Florian Oehler, Stefan Förster, Michael Bron; Higher-Valent Nickel Oxides with Improved Oxygen Evolution Activity and Stability in Alkaline Media Prepared by High-Temperature Treatment of $Ni(OH)_2$. *ACS Catal.* 20 March 2020; 10 (6): 3595–3603.

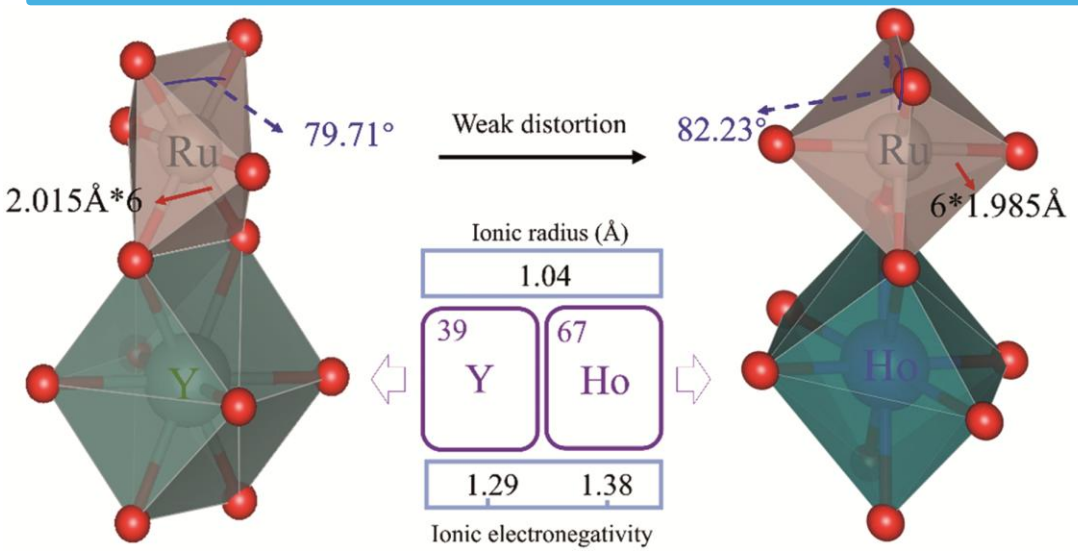
After Pt doped, Tensile strain improves lattice oxygen stability, while local compressive strain enhances OER activity by weakening the adsorption of oxygen intermediates.



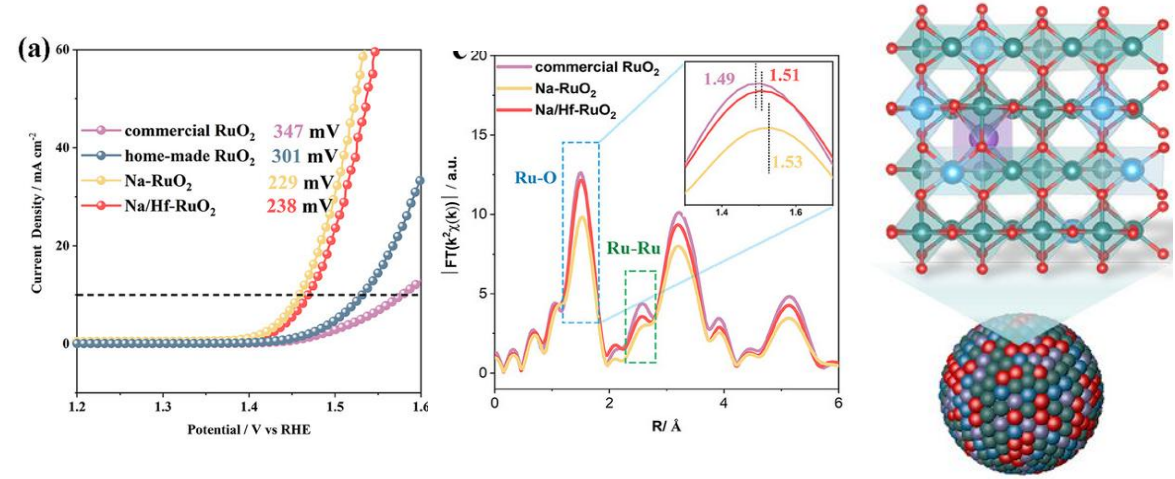
Cao, X., Miao, L., Jia, W. *et al.* Strain heterogeneity in RuO_2 for efficient acidic oxygen evolution reaction in proton exchange membrane water electrolysis. *Nat Commun* **16**, 6217 (2025). https://doi.org/10.1038/s41467-025-58570-3

3.6 Similarities

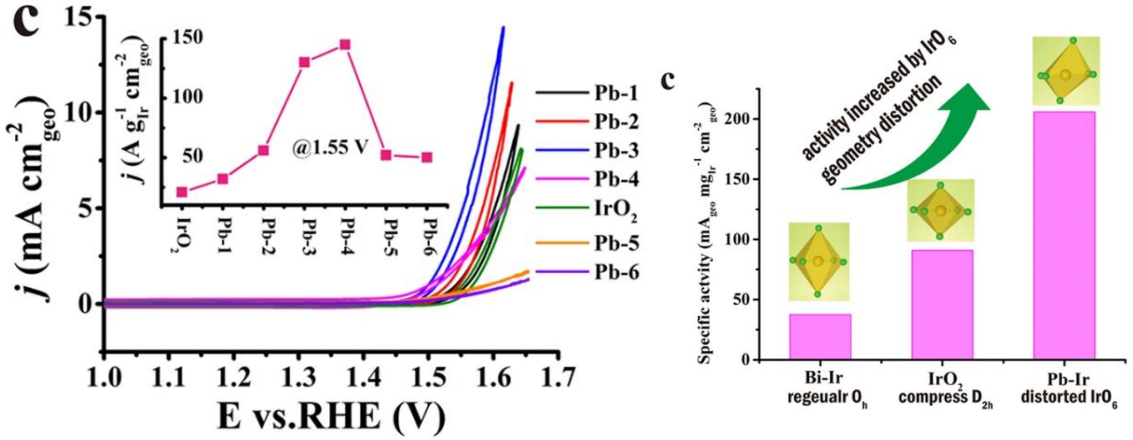
These methods improve OER activity by changing the M–O bond lengths or O–M–O bond angles at active sites, which modifies the local geometry and optimizes the adsorption strength of OER intermediates.



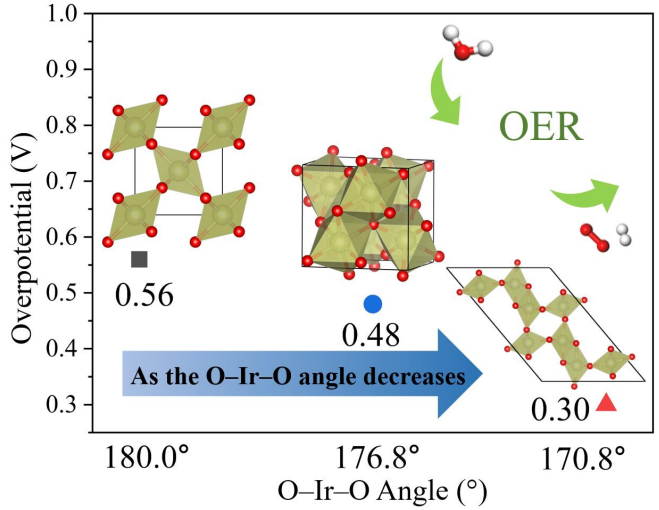
Yan, J.; Zhu, J.; Chen, D.; et al. Mitigation of RuO_6 Octahedron Distortion by Enhanced A-Site Electronegativity in Pyrochlore for Acidic Water Oxidation. J. Mater. Chem. A 2022, 10 (17), 9419–9426.



Zhu, Y.; Wu, F.; Zhang, X.; et al. Engineering Lattice Distortion in Ruthenium Oxide Enables Robust Acidic Water Oxidation via Direct O–O Coupling. Adv. Mater. 2025, 37, 2500449.

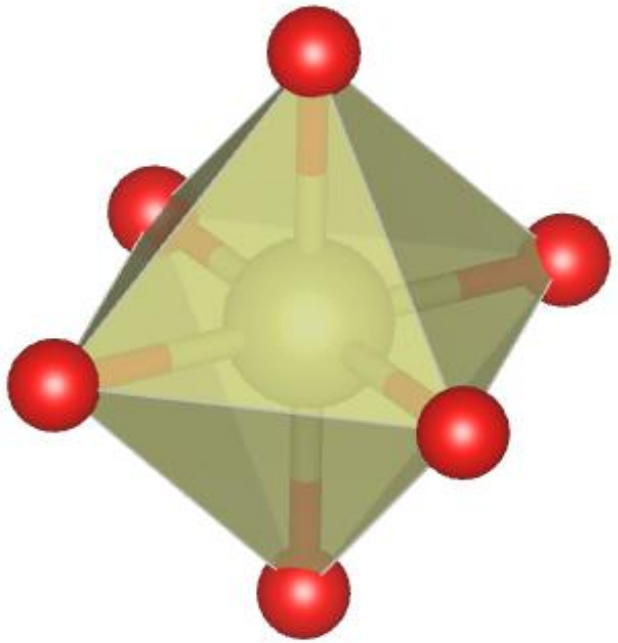


Sun, W.; Liu, J.-Y.; Gong, X.-Q.; et al. OER Activity Manipulated by IrO_6 Coordination Geometry: An Insight from Pyrochlore Iridates. Sci. Rep. 2016, 6, 38429.



Wu, Y.; Thakur, N.; Kumar, M.; et al. Langmuir 2026, 42 (24), 17541–17552.

3.7 Motivation

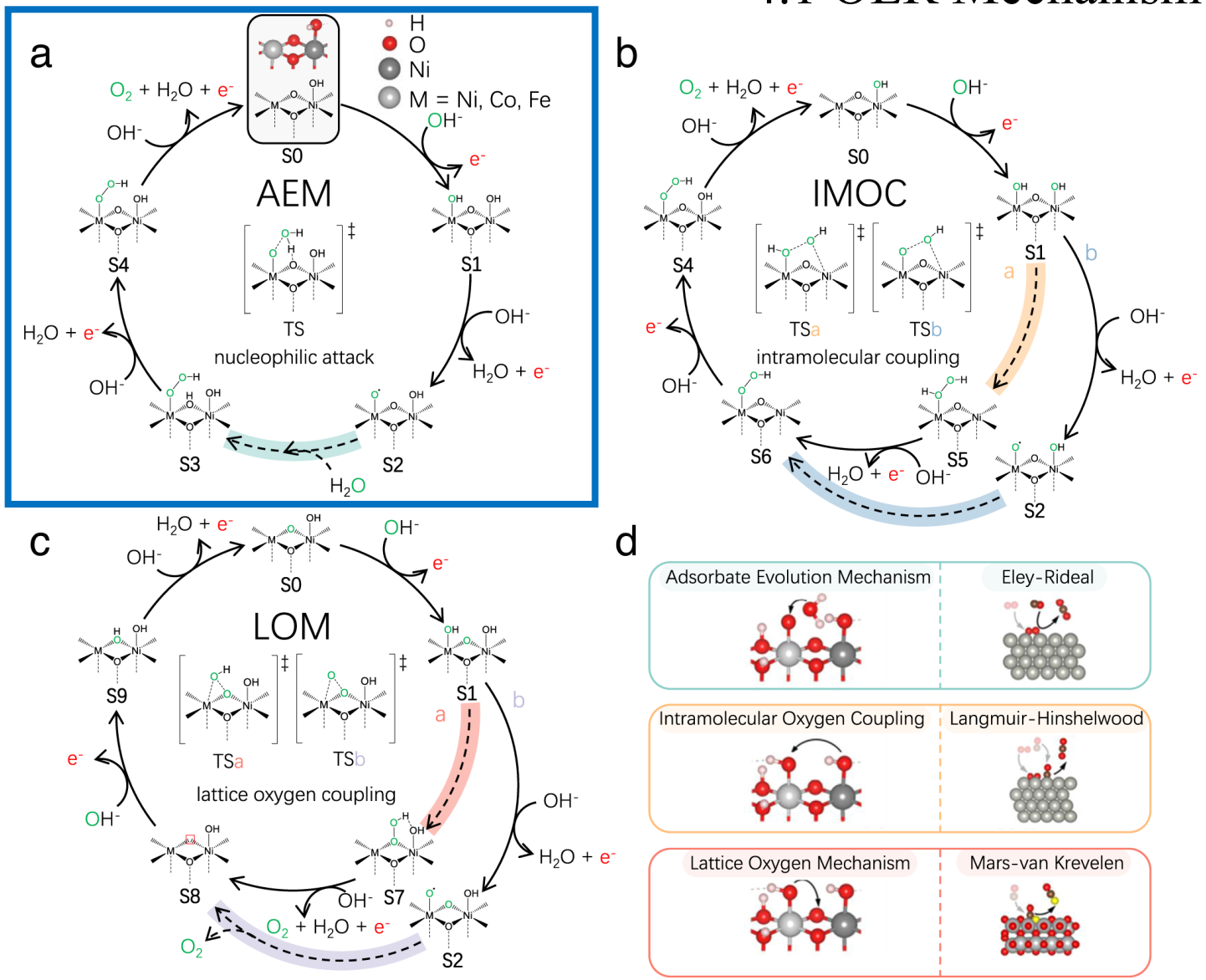


Which geometric parameters in the crystal lattice of six coordinated metal oxides (MO_6) can be tuned to improve OER activity?

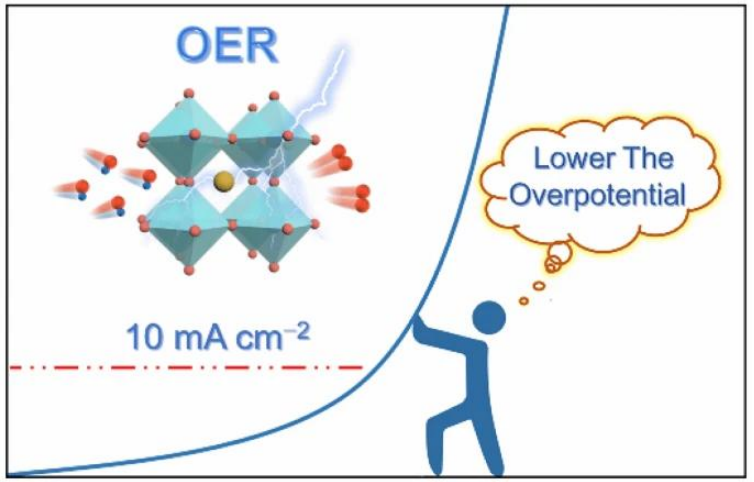
Bond length…
Or
Bond angle…

4. Research Methods and Theoretical Foundations

4.1 OER Mechanism



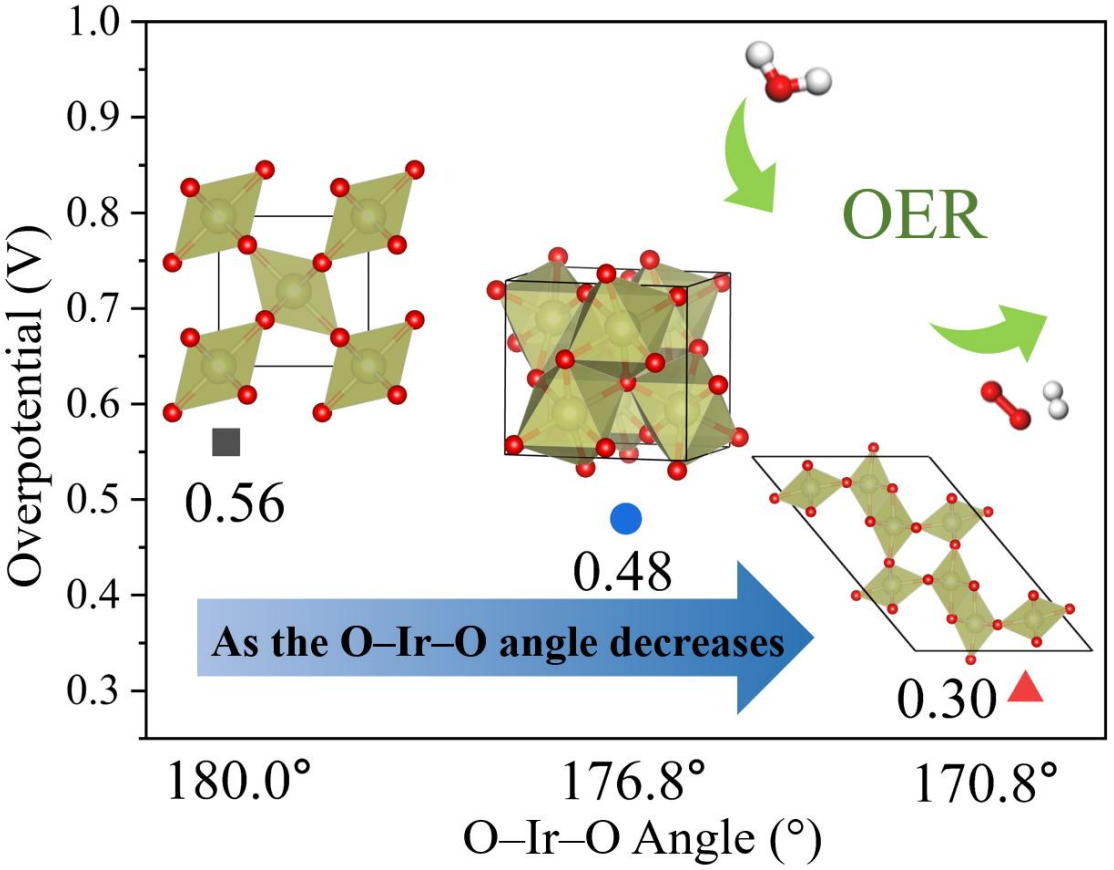
A lower overpotential indicates higher OER catalytic activity.



Improving OER activity requires adjusting the adsorption of intermediates, either by strengthening or weakening their adsorption.

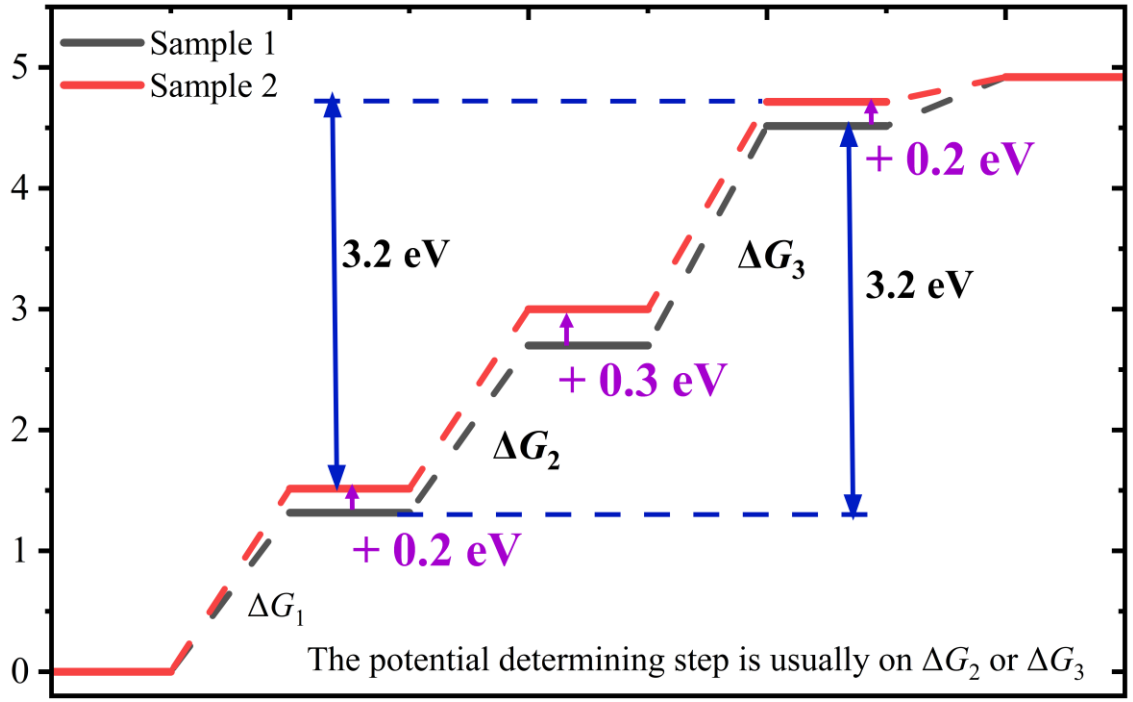
Wang, Z., Goddard, W.A. & Xiao, H. Potential-dependent transition of reaction mechanisms for oxygen evolution on layered double hydroxides. *Nat Commun* **14**, 4228 (2023).

4.2 Theoretical Findings on Iridium Oxides



Changes in crystal structure (O-Ir-O) affect the bonding interactions between active sites and OER intermediates, thereby changing the adsorption energy.

How do bond angles affect OER activity?



$$\Delta G_{2\text{-}S1} + \Delta G_{3\text{-}S1} = \Delta G_{2\text{-}S2} + \Delta G_{3\text{-}S2} = 3.2 \text{ eV}$$

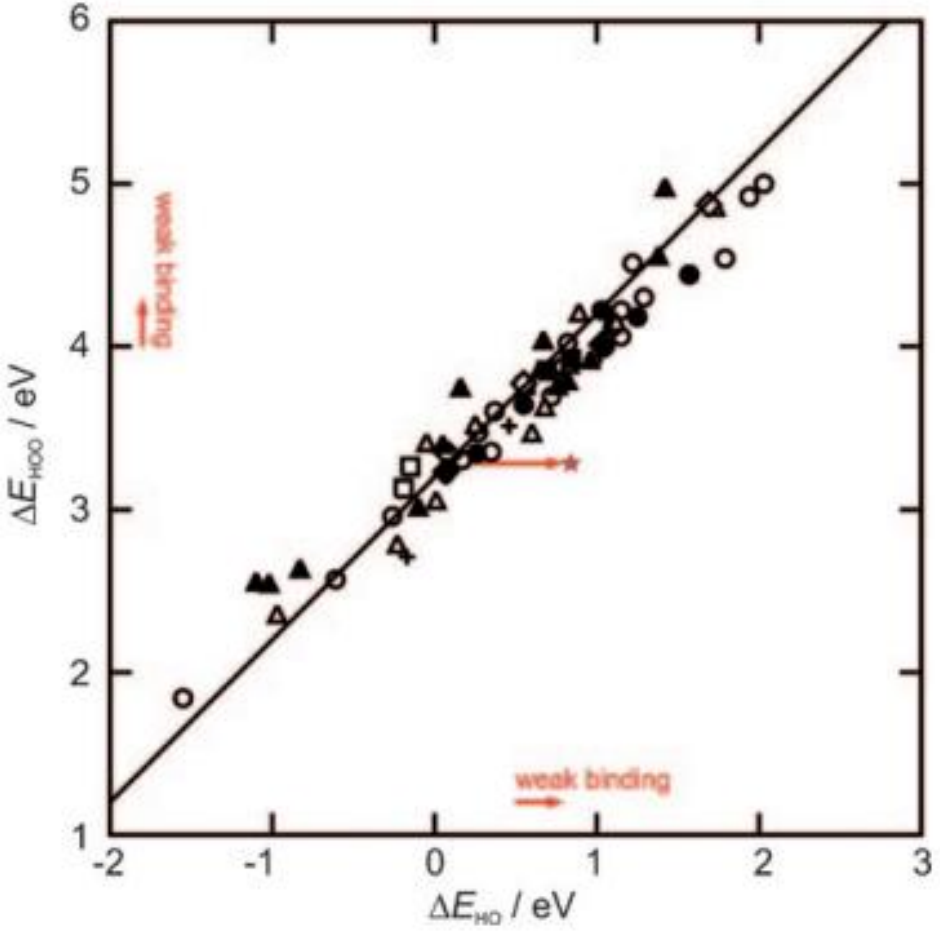
$$\Delta G_{2\text{-}S1} > \Delta G_{2\text{-}S1} \rightarrow \Delta G_{3\text{-}S2} < \Delta G_{3\text{-}S1}$$

4.3 Single Metal Atom Calculation

OER activity trends in transition metal oxides

Difference of ΔE_{HOO*}- ΔE_{HO*} is 3.2 eV for metal oxides surfaces and metal surfaces.

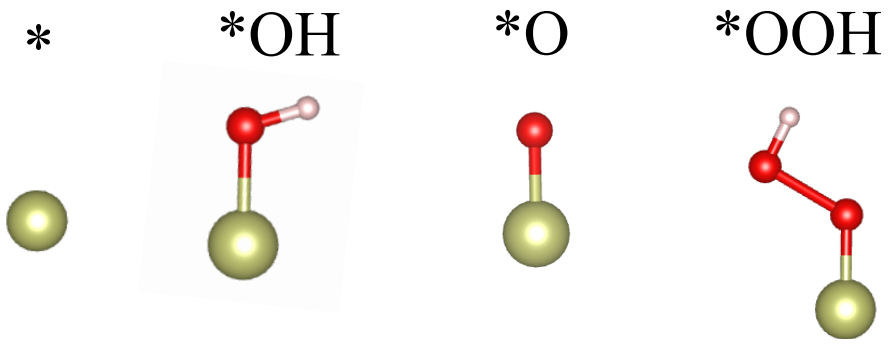
$$\Delta E_{HOO*} = \Delta E_{HO*} + 3.20 \text{ eV}.[1]$$

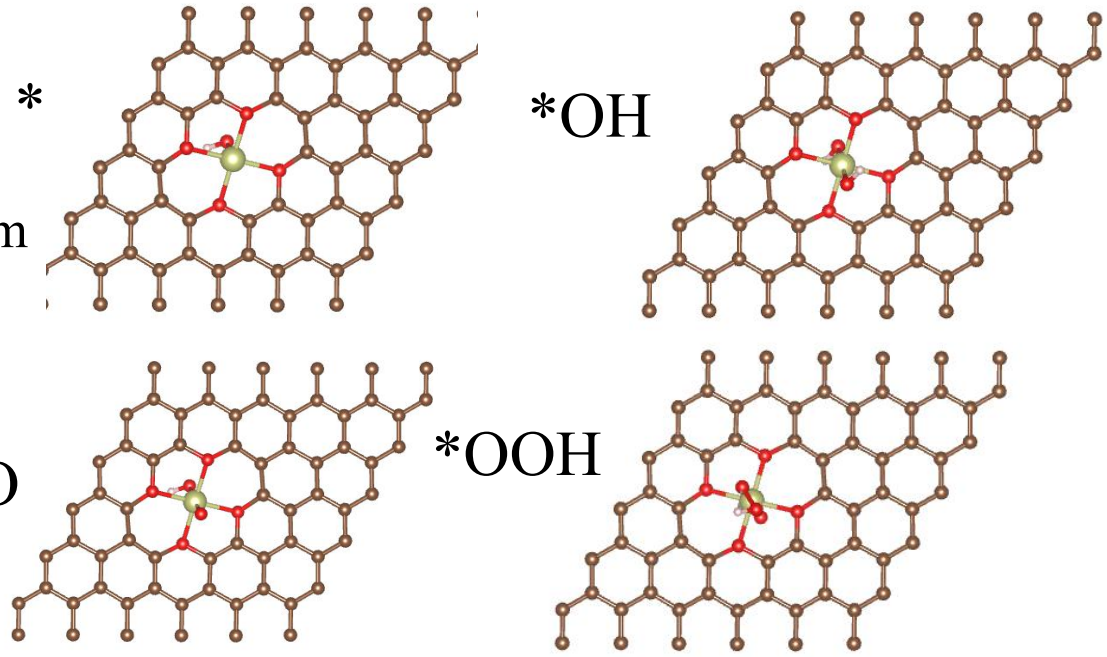


Can this trend also be observed in other structures, rather than only at the surface?

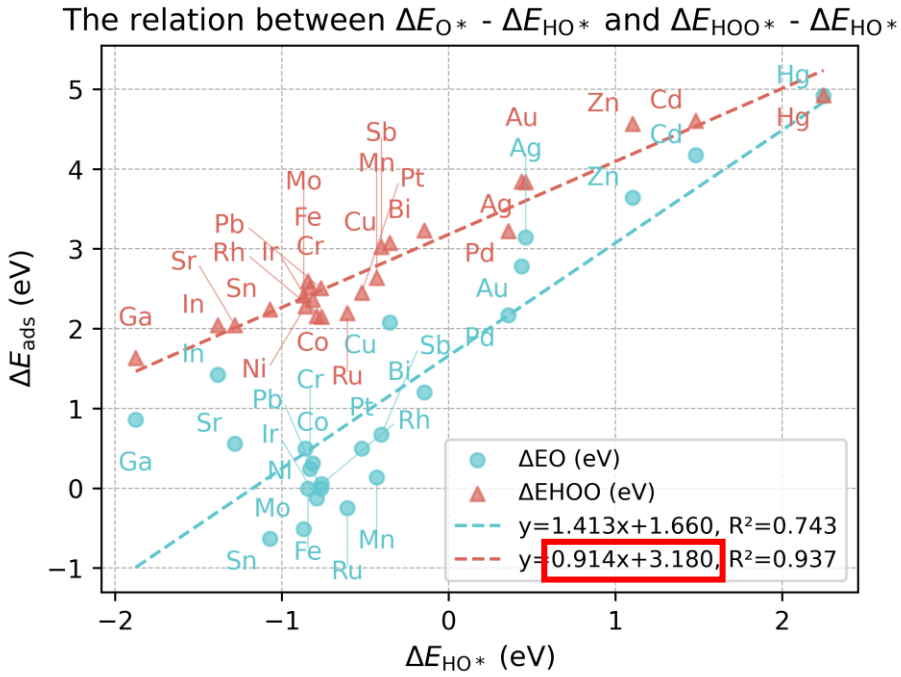
[1] Man, I, C.; Su, H, Y.; Calle-Vallejo, F. et al. *ChemCatChem*, **2011**, *3*(7): 1159-1165.

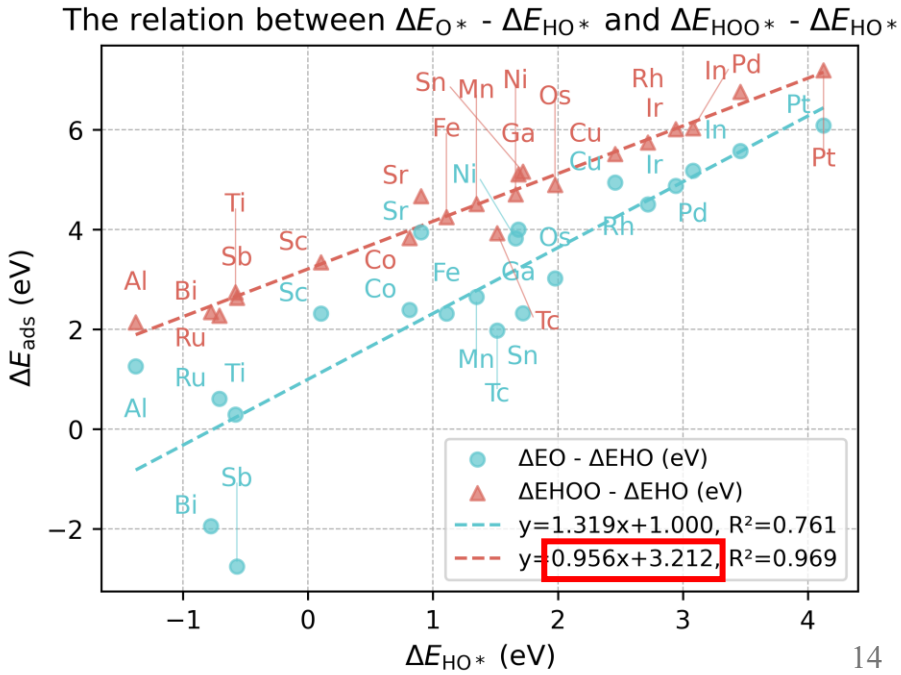
4.4 Universal Scaling Relationship





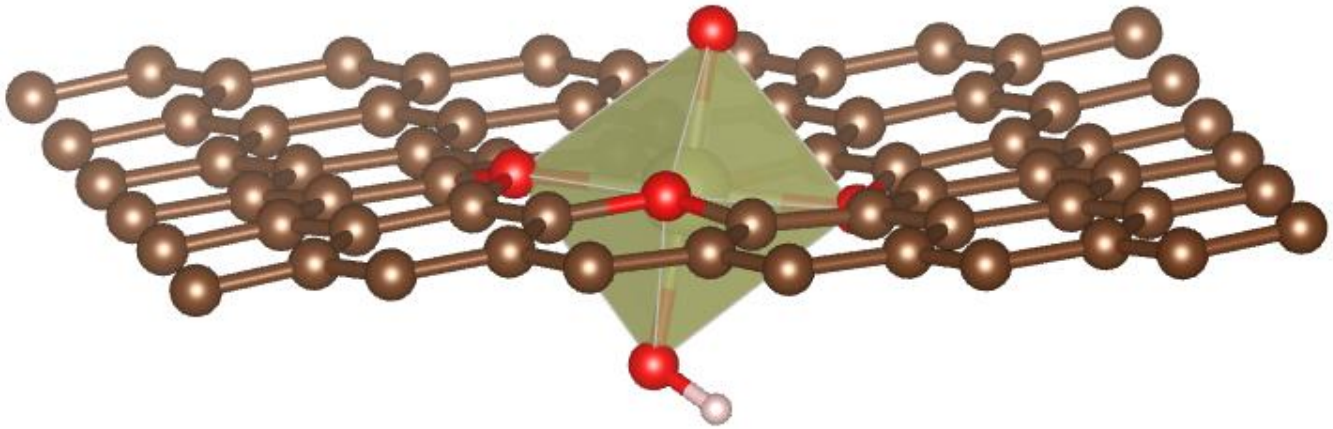
Single-atom models, single-atom catalysts, metal oxides, and metal surfaces generally follow the same relationship, where $\Delta E_{HOO*} - \Delta E_{HO*} \approx 3.2$ eV.



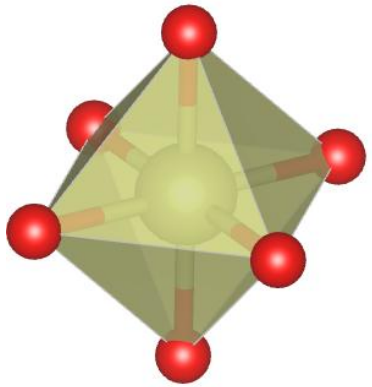


Model selection: Why SACs on graphene?

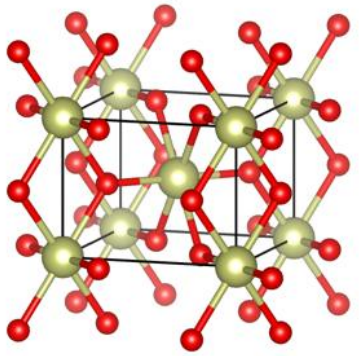
Can graphene be used as a platform for tuning O–M–O bond angles?



- More stable than isolated clusters

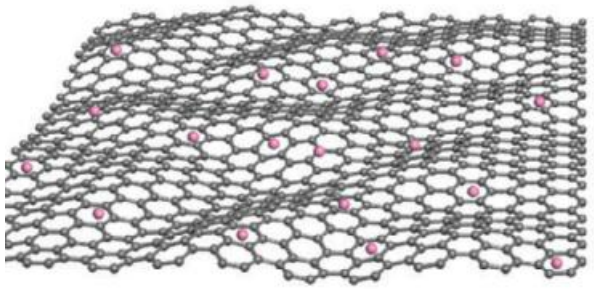


- Easier to adjust bond angles than in bulk metal oxides



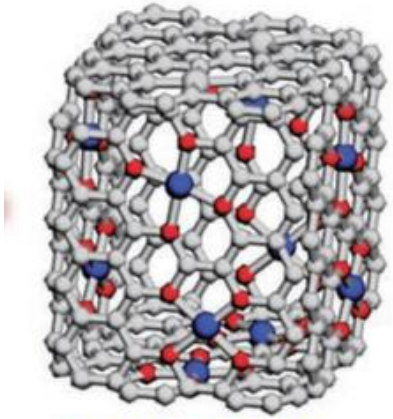
4.5 Oxygen Coordinated Single-Atom Catalysts

SnO_6 Single-Atom Catalyst



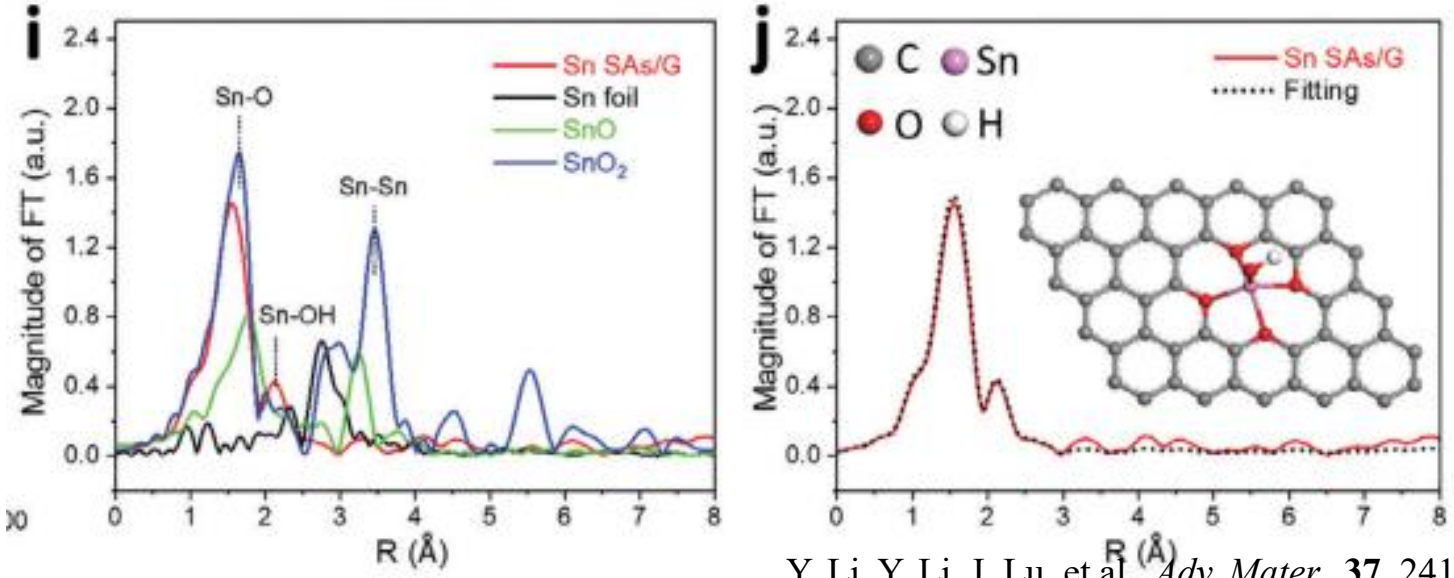
Sn SAs/G

NiO_6 Single-Atom Catalyst

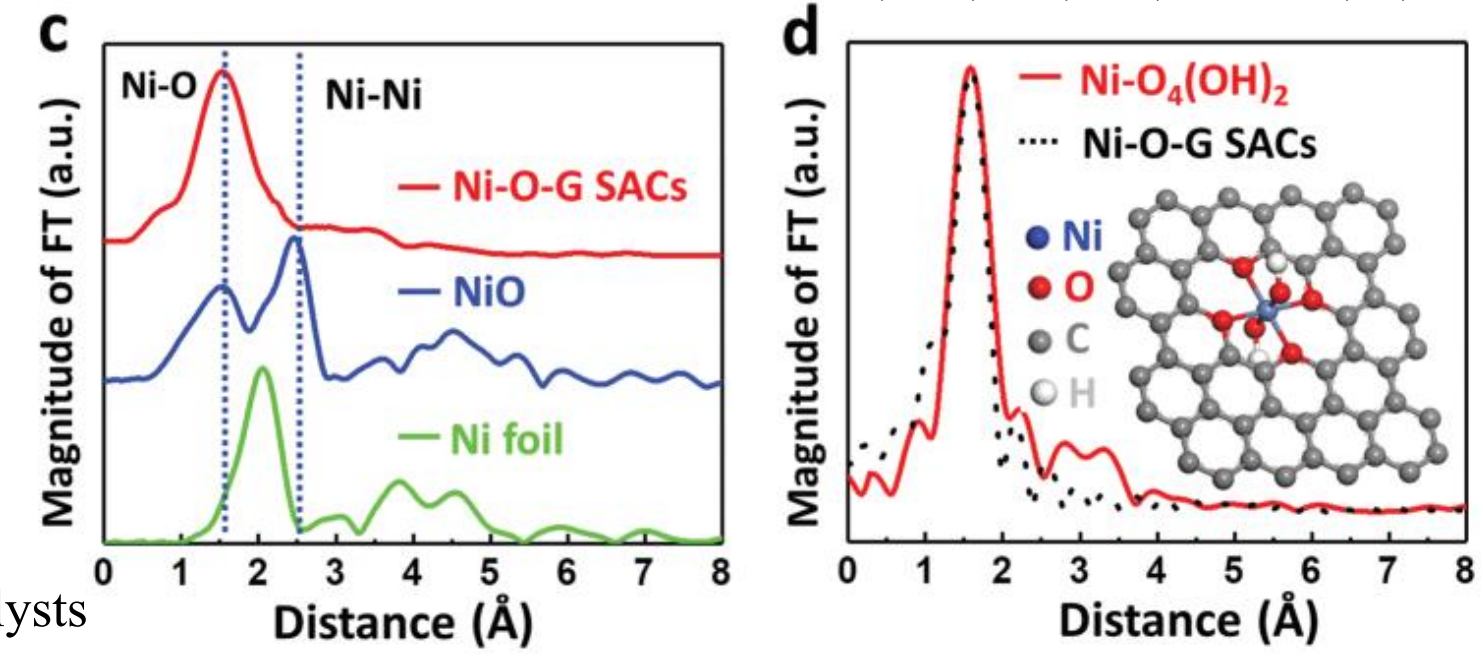


Ni-O-G SACs

Oxygen-coordinated single-atom catalysts can be experimentally synthesized.



Y. Li, Y. Li, J. Lu, et al., *Adv. Mater.*, **37**, 2415026 (2025).



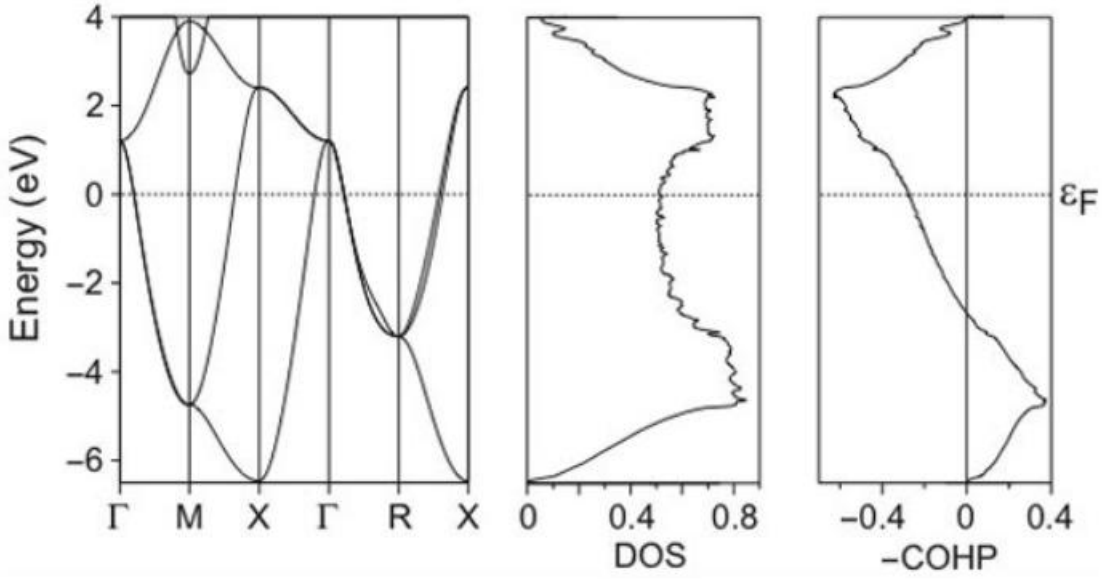
Y. Li, Z. S. Wu, P. Lu, et al., *Adv. Sci.*, **7**, 1903089 (2020).

5. Bonding Interaction and OER

5.1 Bond analysis

Crystal Orbital Hamilton Population (**COHP**)

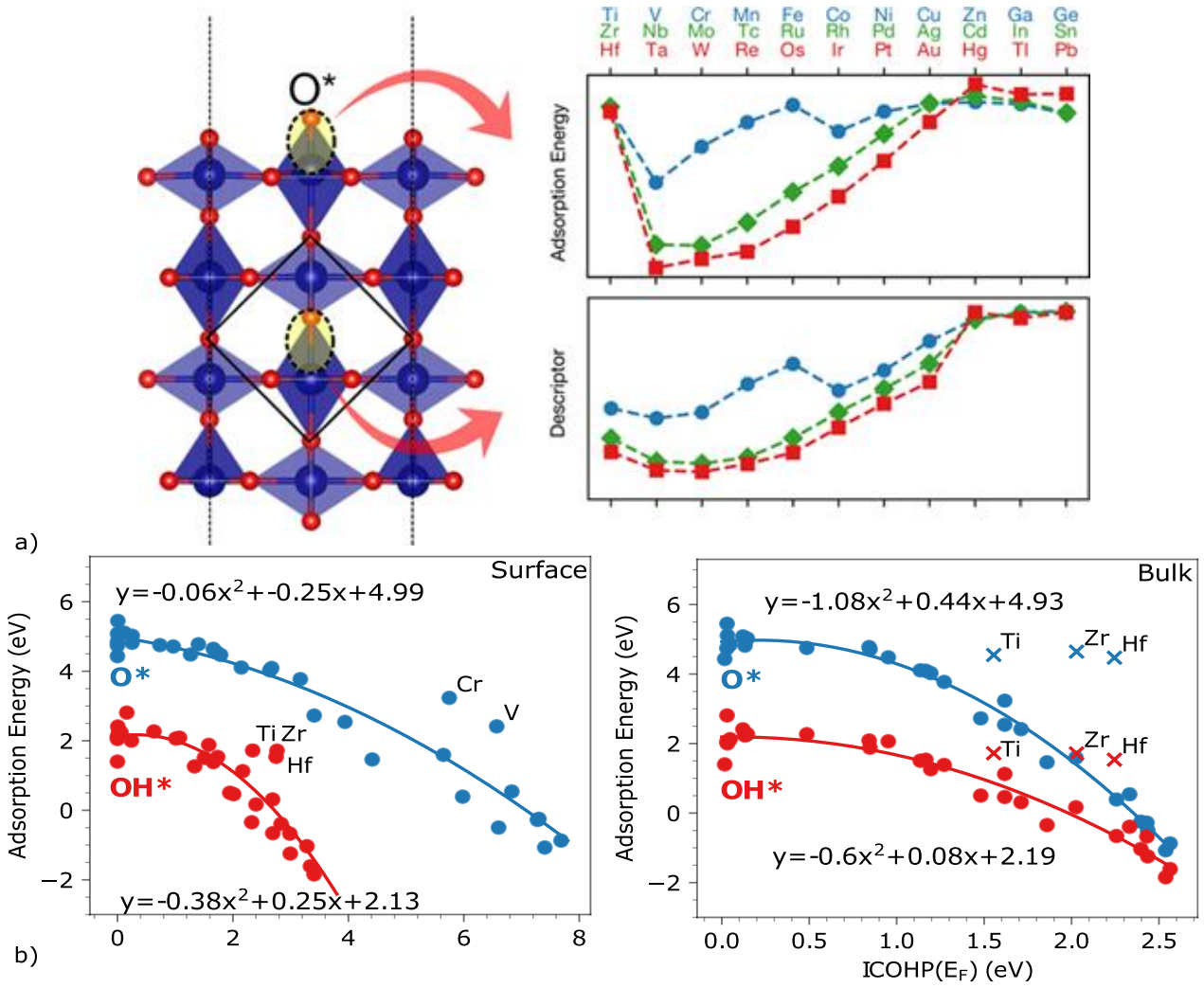
Bonding Interaction

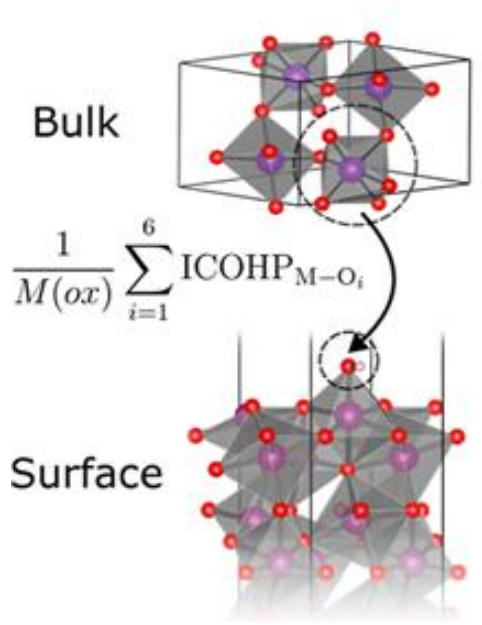


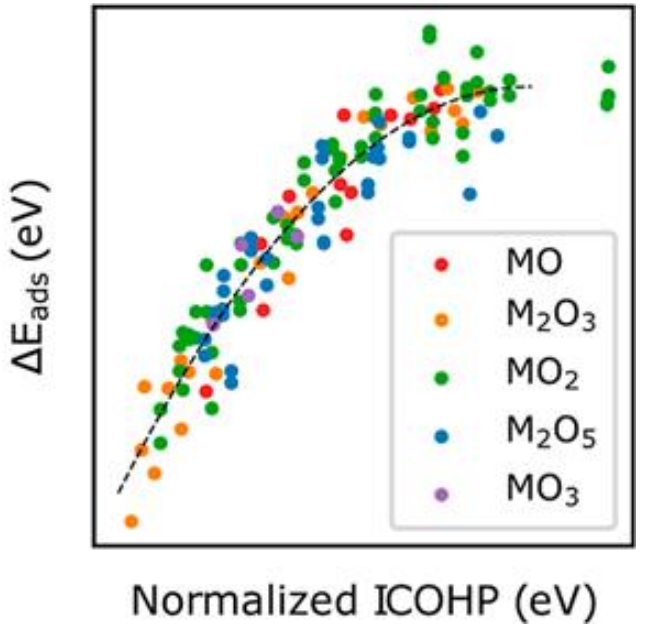
COHP calculations can quickly evaluate bonding interaction, but how can this be related to adsorption energy in OER?

Nelson, R.; Ertural, C.; George, J.; et al. LOBSTER: Local Orbital Projections, Atomic Charges, and Chemical-Bonding Analysis from Projector-Augmented-Wave-Based Density-Functional Theory. J. Comput. Chem. 2020, 41 (21), 1931–1940.

5.2 Relationship Between Bonding Interaction and Adsorption Energy







When ICOHP is between approximately −4.0 and −2.5 eV, it shows an approximately linear relationship with the adsorption energies of OH* and O* groups.

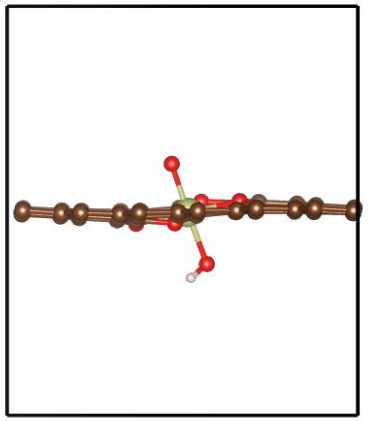
[1] Comer, B. M.; Li, J.; Abild-Pedersen, F.; Bajdich, M.; Winther, K. T. Unraveling Electronic Trends in O* and OH* Surface Adsorption in the MO_2 Transition-Metal Oxide Series. J. Phys. Chem. C 2022, 126 (18), 7903–7909.
[2] Comer, B. M.; Bothra, N.; Lunger, J. R.; Abild-Pedersen, F.; Bajdich, M.; Winther, K. T. Prediction of O and OH Adsorption on Transition Metal Oxide Surfaces from Bulk Descriptors. ACS Catal. 2024, 14 (7), 5286–5296.

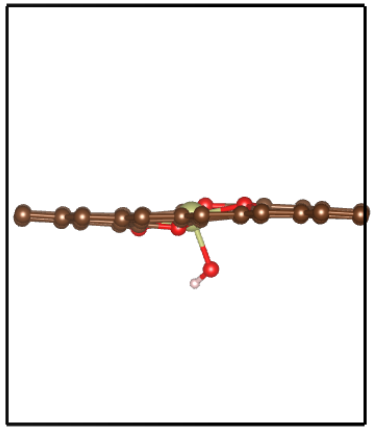
5. 3 Validation

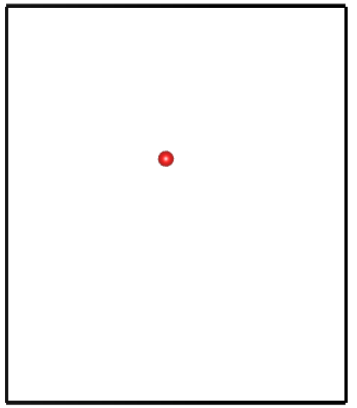
Calculation of adsorption energies using single-point DFT calculations on SACs

$$E_{\text{ads}} = E_{\text{full}} - E_{\text{full without group}} - E_{\text{group}}$$

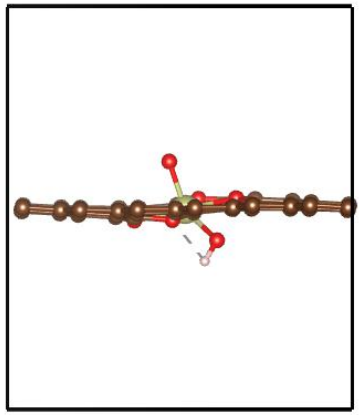
O–M–O : 180°

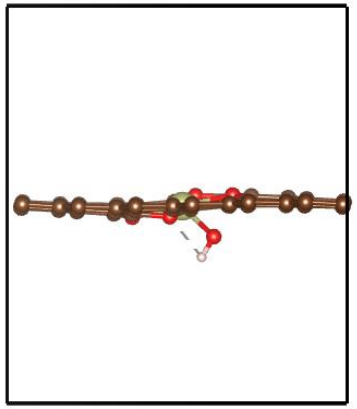


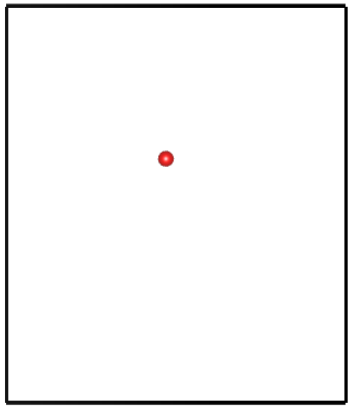




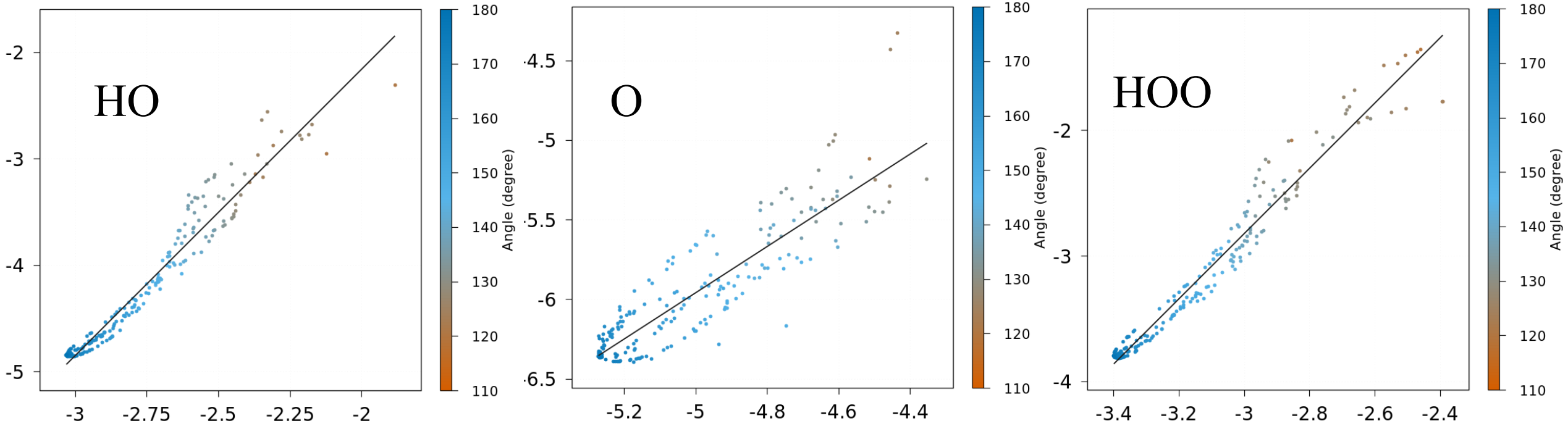
O–M–O : 125°







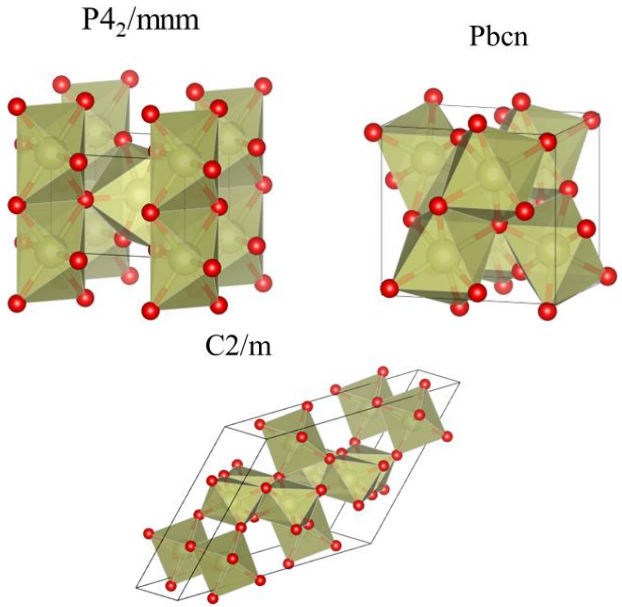
5.4 Validation of the linear relationship between COHP and adsorption energy on SACs with IrO_6



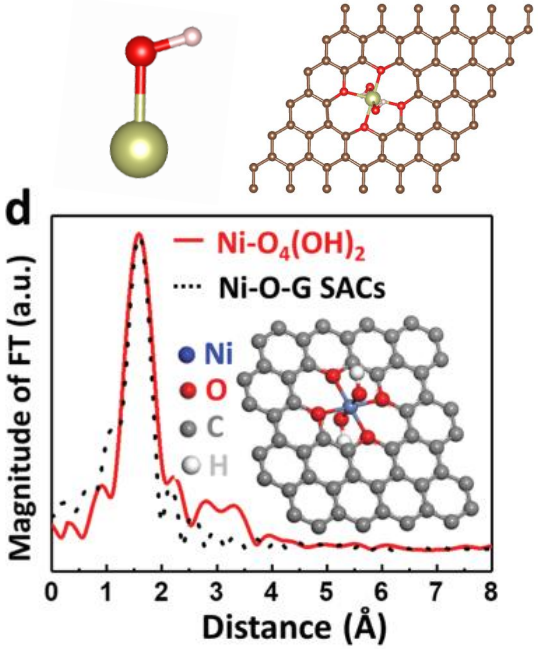
- The strongest linear correlation occurs at O–M–O angles of 140° ~ 180°.
- IpCOHP and adsorption energy show a positive correlation.
- This agrees with previous studies showing a linear relationship for IpCOHP < −2.4 eV.

5.5 Summary of the Methodology

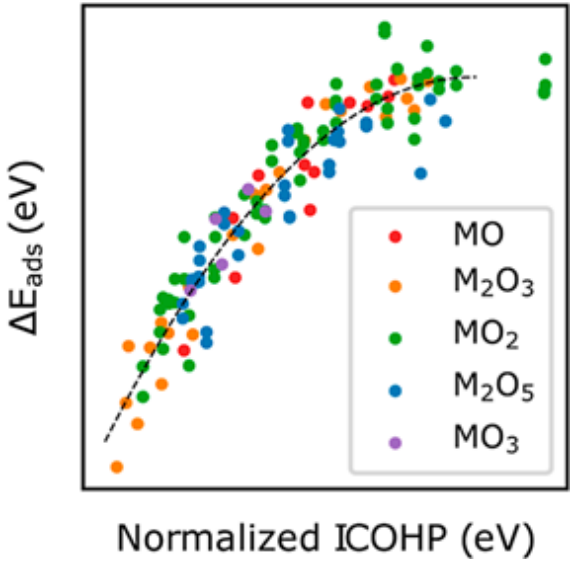
Investigate the effect of O–M–O bond angles on bonding strength using single-point DFT calculations and COHP analysis.



Bond Angle Effects: Bond angles affect bonding strength, but studying their effects in crystal structures is challenging.



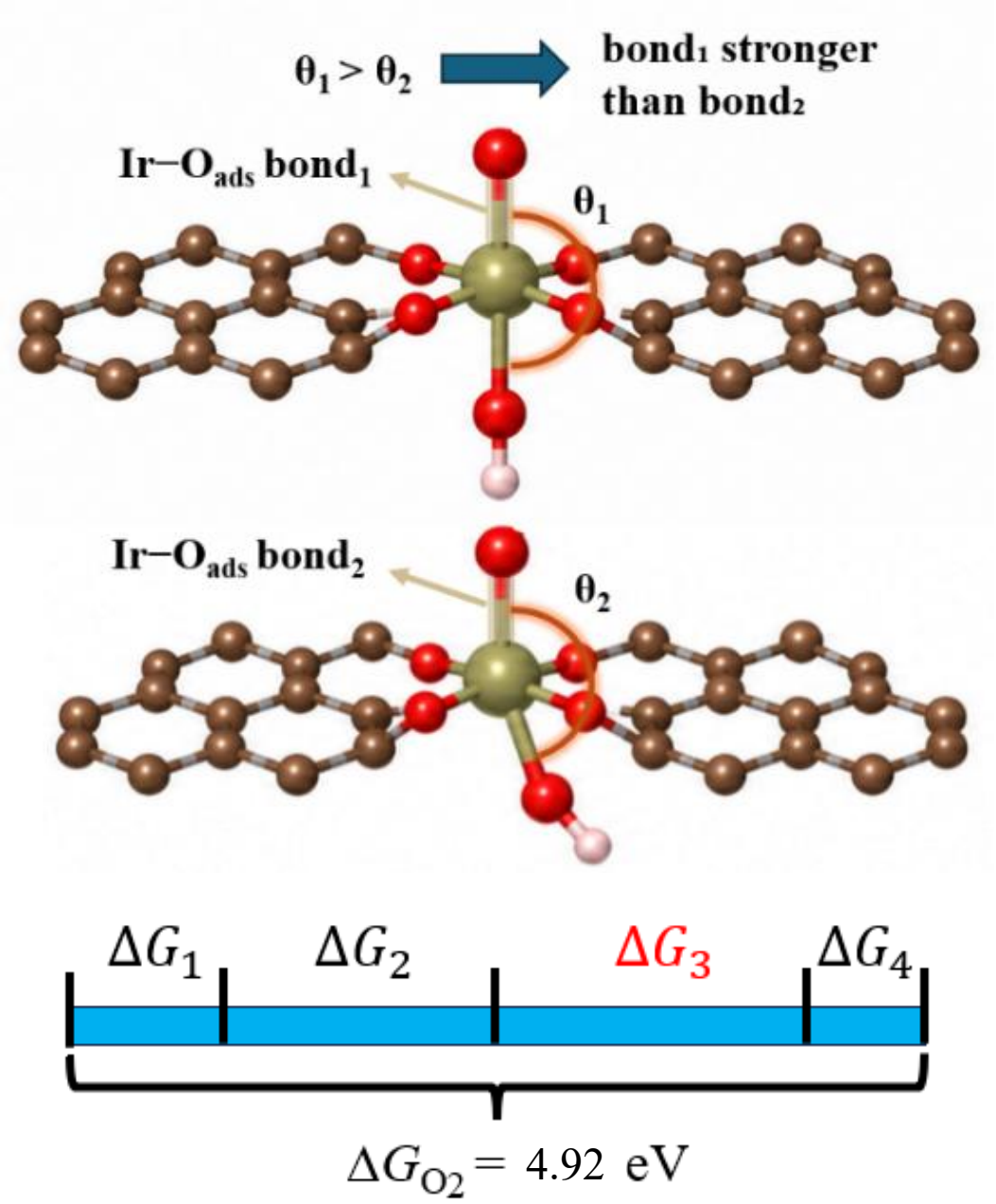
Modeling Possibility: Such coordination structures have been experimentally synthesized, and both simple and complex models show similar trends.



COHP–ΔE_{ads} Relationship: Bonding strength descriptors, such as ICOHP, can be correlated with adsorption energies.

Use IpCOHP trends as a descriptor to predict adsorption energy trends across different groups.

5.6 Method Abstract

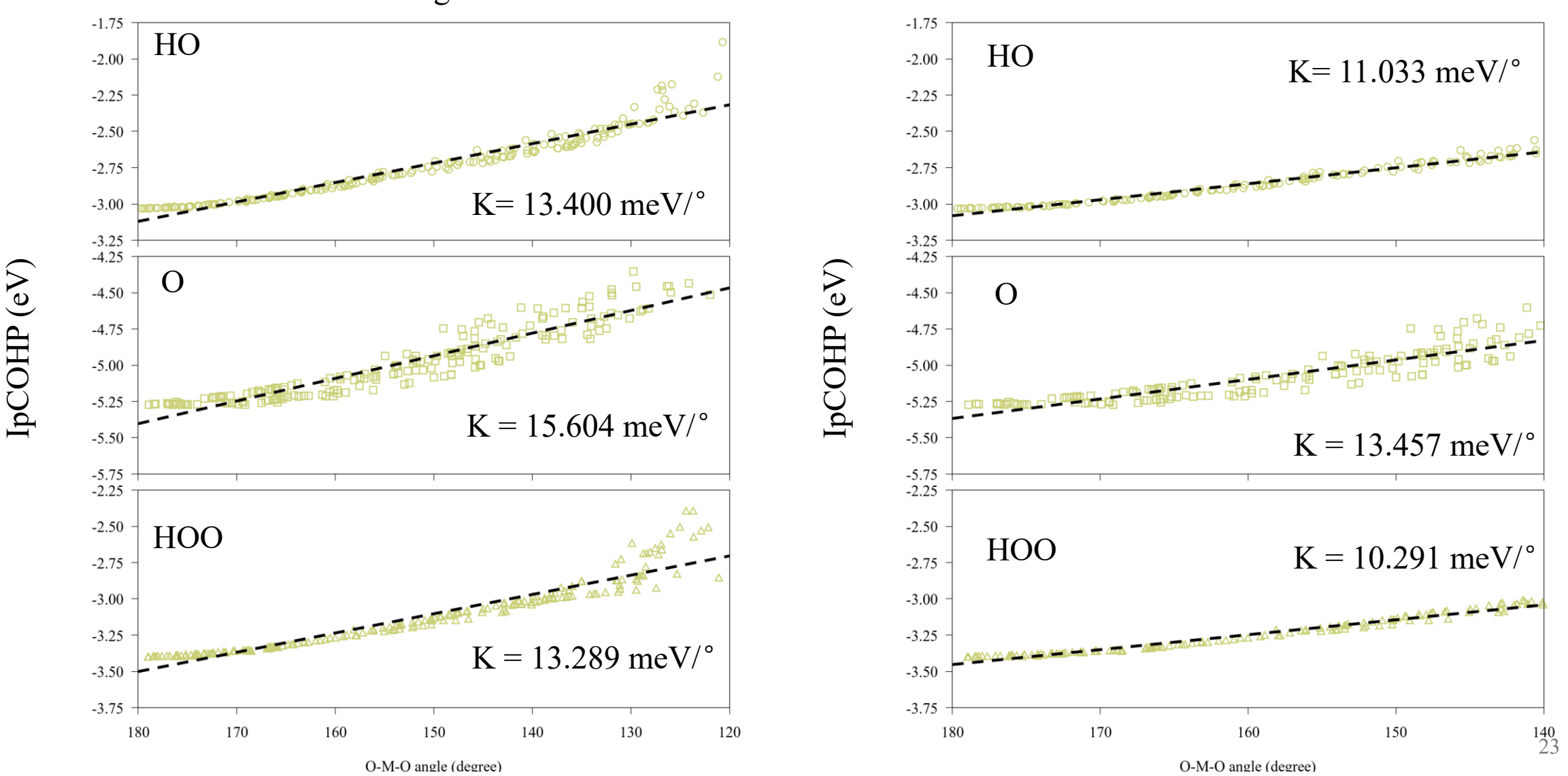


- Generate 1,200 structures with O–M–O bond angles ranging from 120° to 180°. Randomly select 200 structures.
- Single-point DFT calculations to obtain IpCOHP values for the upper M–O bonds.
- Calculate the slope of IpCOHP with respect to the bond angle. Compare the slopes across three different groups.
- Analyze the effect of bond angles on OER activity based on the IpCOHP slopes.
- Use the O–M–O bond angles of graphene-supported MO_6 structures as a model for angle changes in six-coordinated metal oxide crystals to OER activity.

6. Variation of IpCOHP with O–M–O bond angle

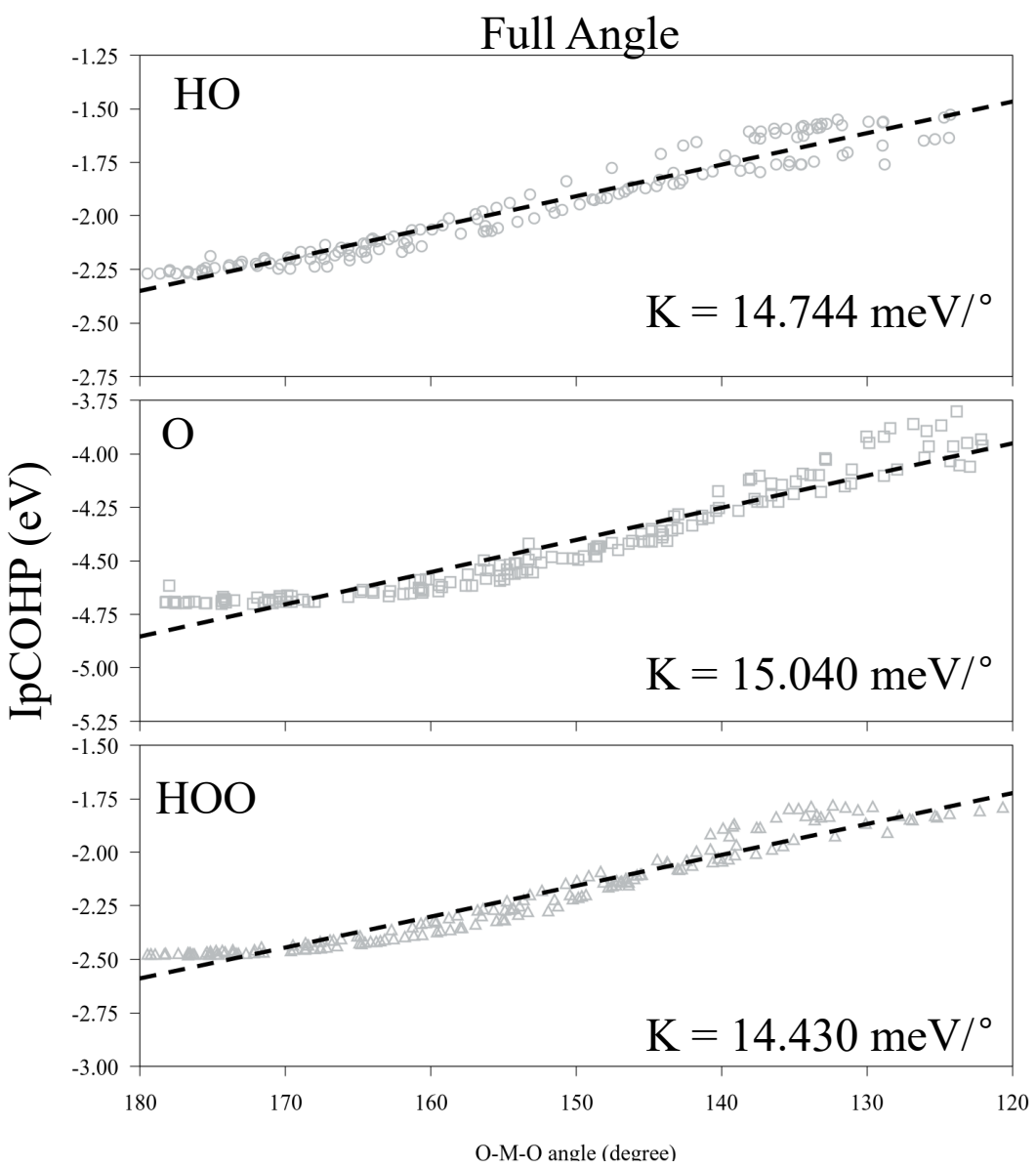
6.1 IrO_6

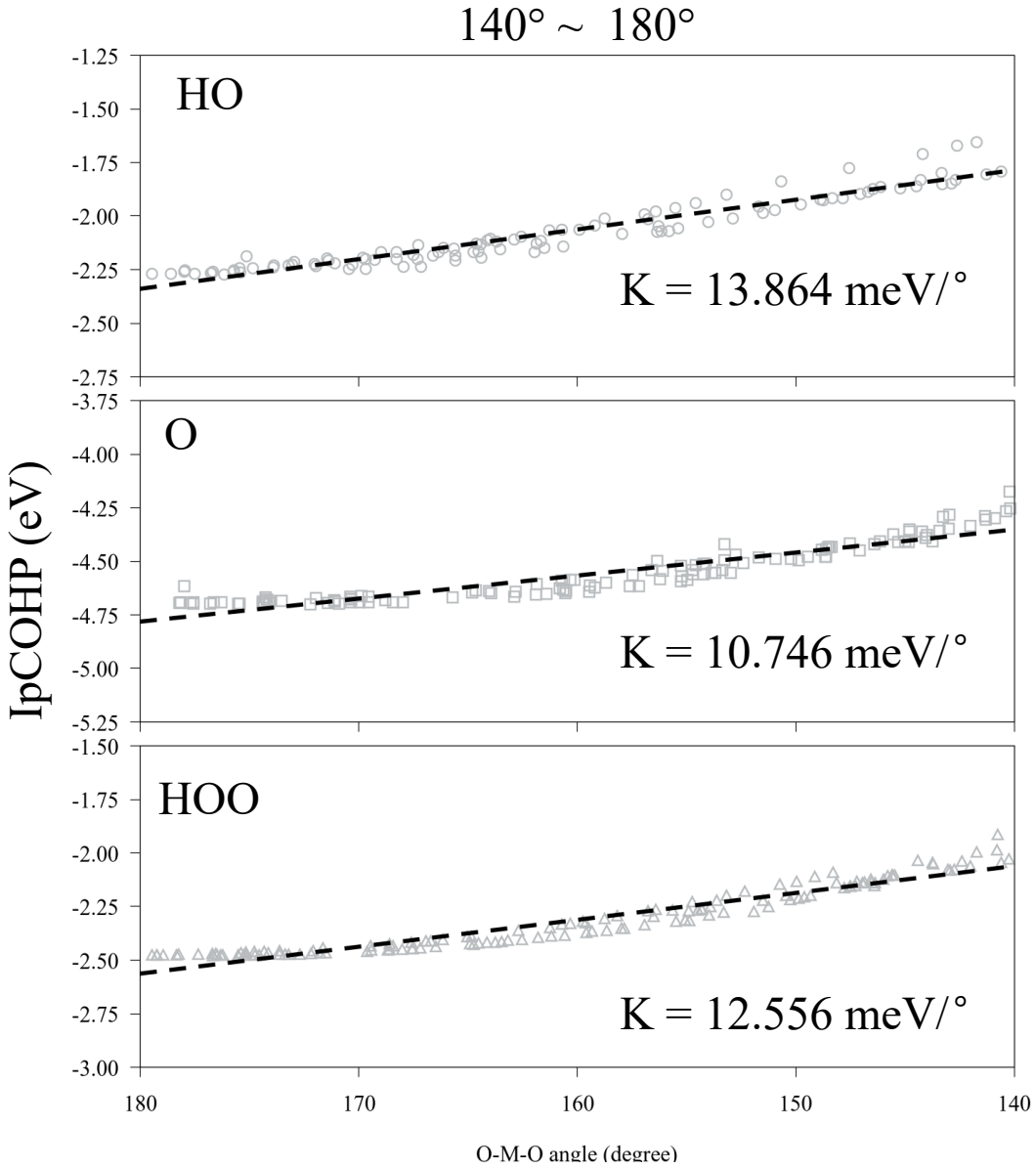
Slope: O > HO ≈ HOO



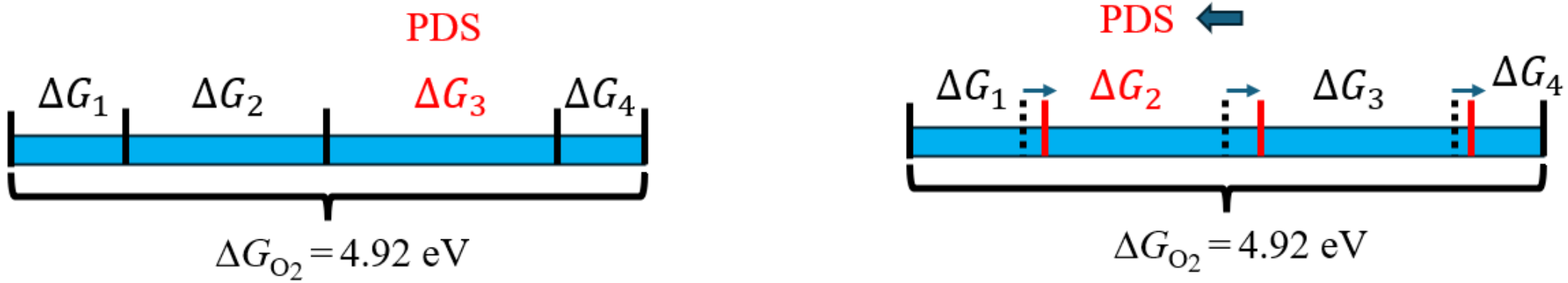
6.2 NiO_6

Slope: O >≈ HO ≈ HOO





For IrO_6 and NiO_6



Expected trends

- HOO stronger than HO → $\Delta G_2 + \Delta G_3$ increases above 3.2 eV.
- HO stronger than HOO → $\Delta G_2 + \Delta G_3$ decreases below 3.2 eV.

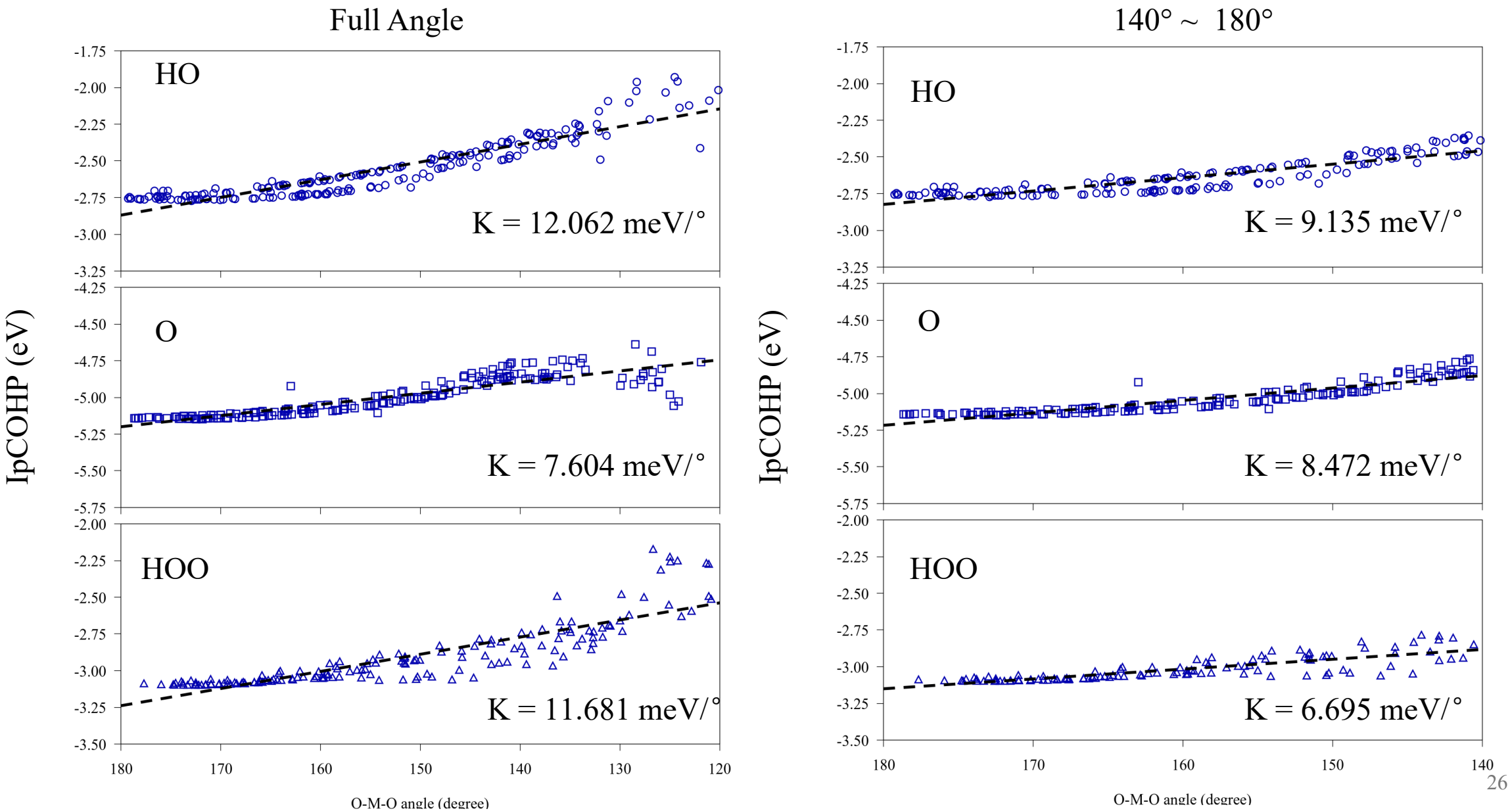
Expected Slopes based on relative rates of change

The bonding interaction between O group and the Ir&Ni atom increases faster

- O changes more than HO and HOO → ΔG_3 decreases, ΔG_2 increases.

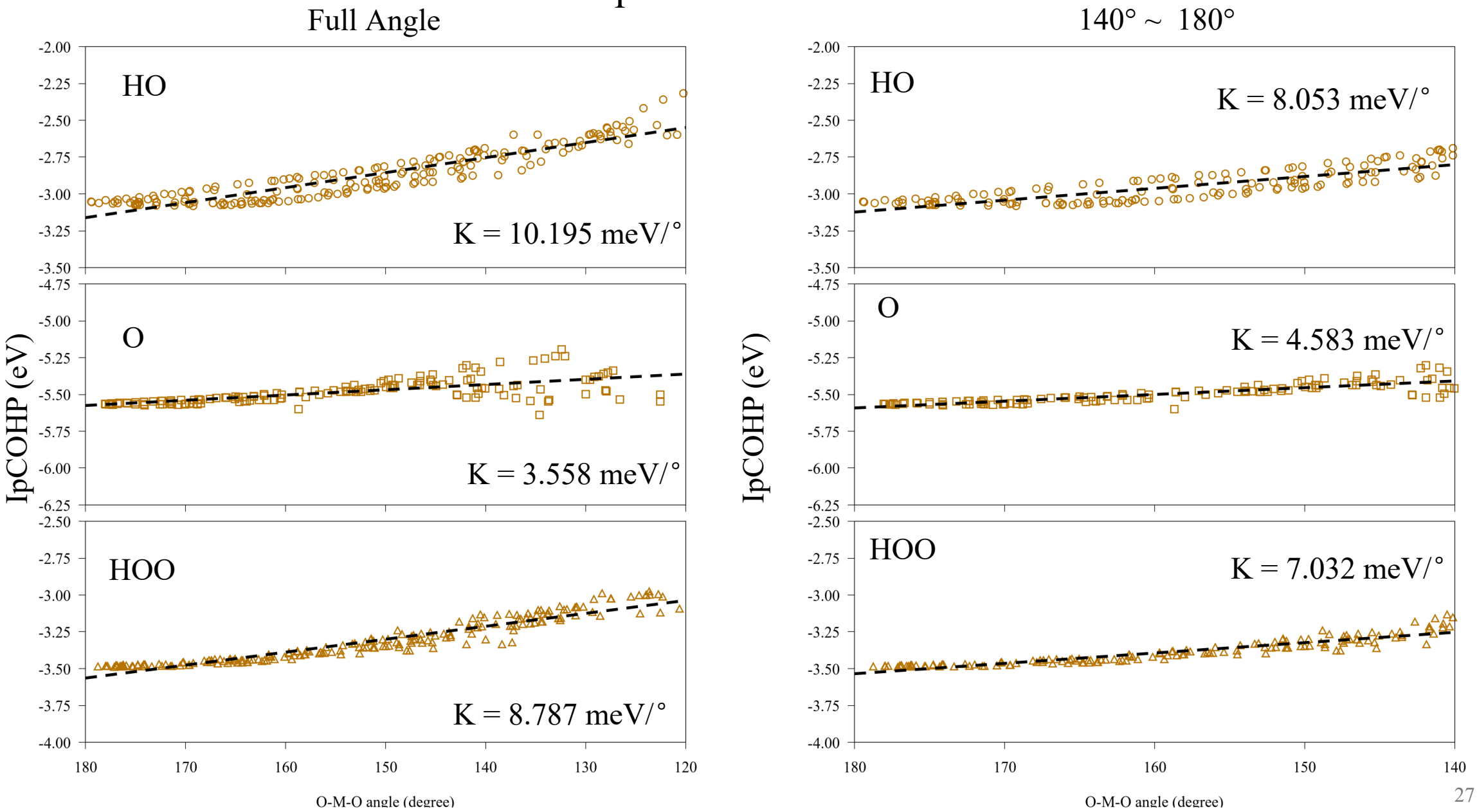
6.3 CoO_6

Slope: HO ≈ HOO > O

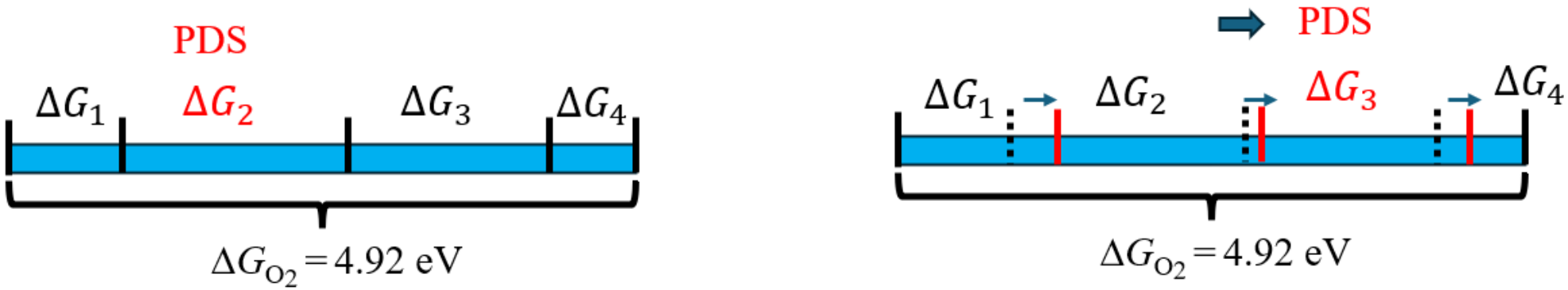


6.4 FeO_6

Slope: HO > HOO > O



For CoO_6 and FeO_6



Expected trends

- HOO stronger than HO → $\Delta G_2 + \Delta G_3$ increases above 3.2 eV.
- HO stronger than HOO → $\Delta G_2 + \Delta G_3$ decreases below 3.2 eV.

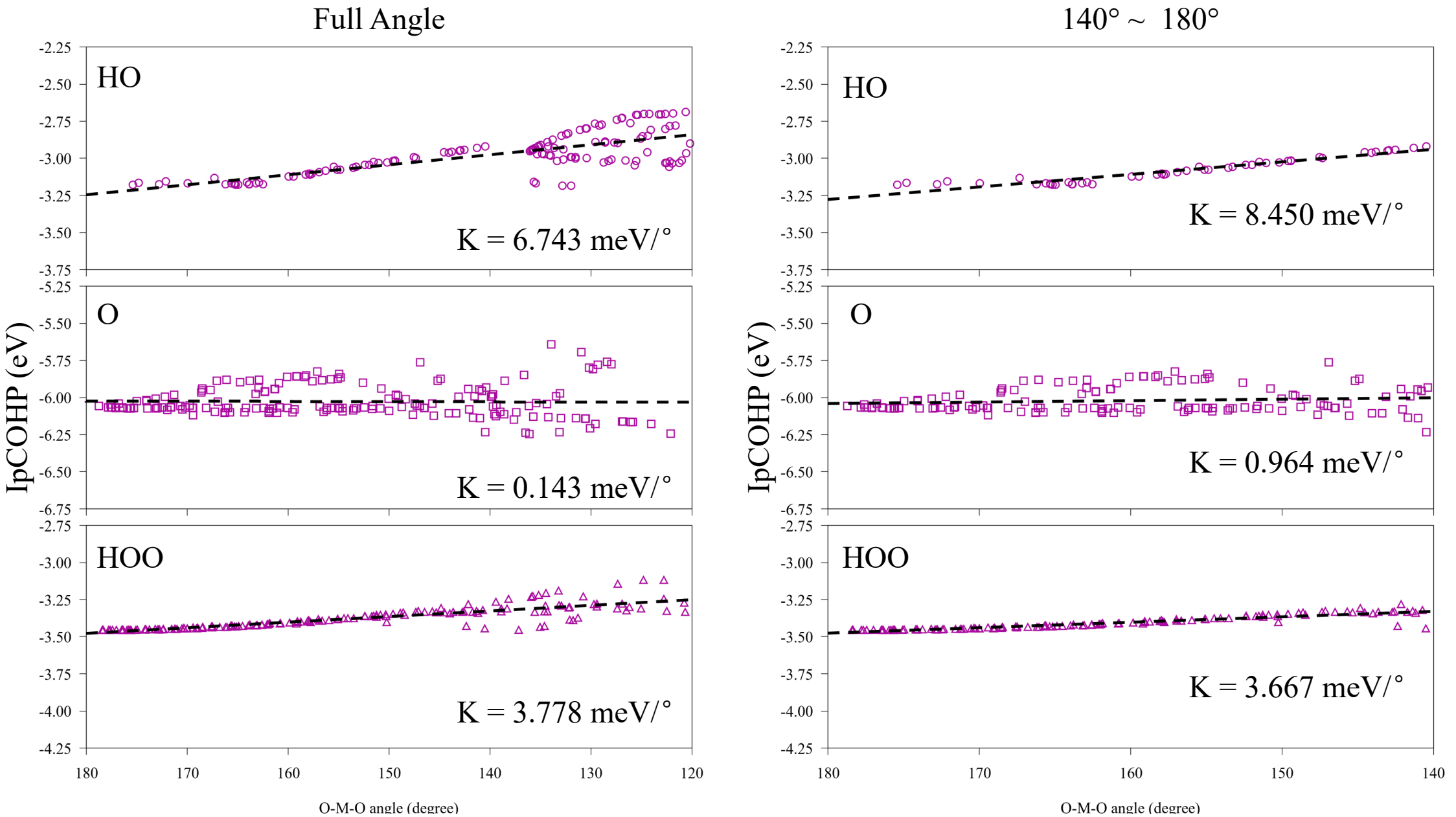
Expected Slopes based on relative rates of change

The bonding interaction between O group and the Co&Fe atom increases slower

- O changes less than HO and HOO → ΔG_2 decreases, ΔG_3 increases.

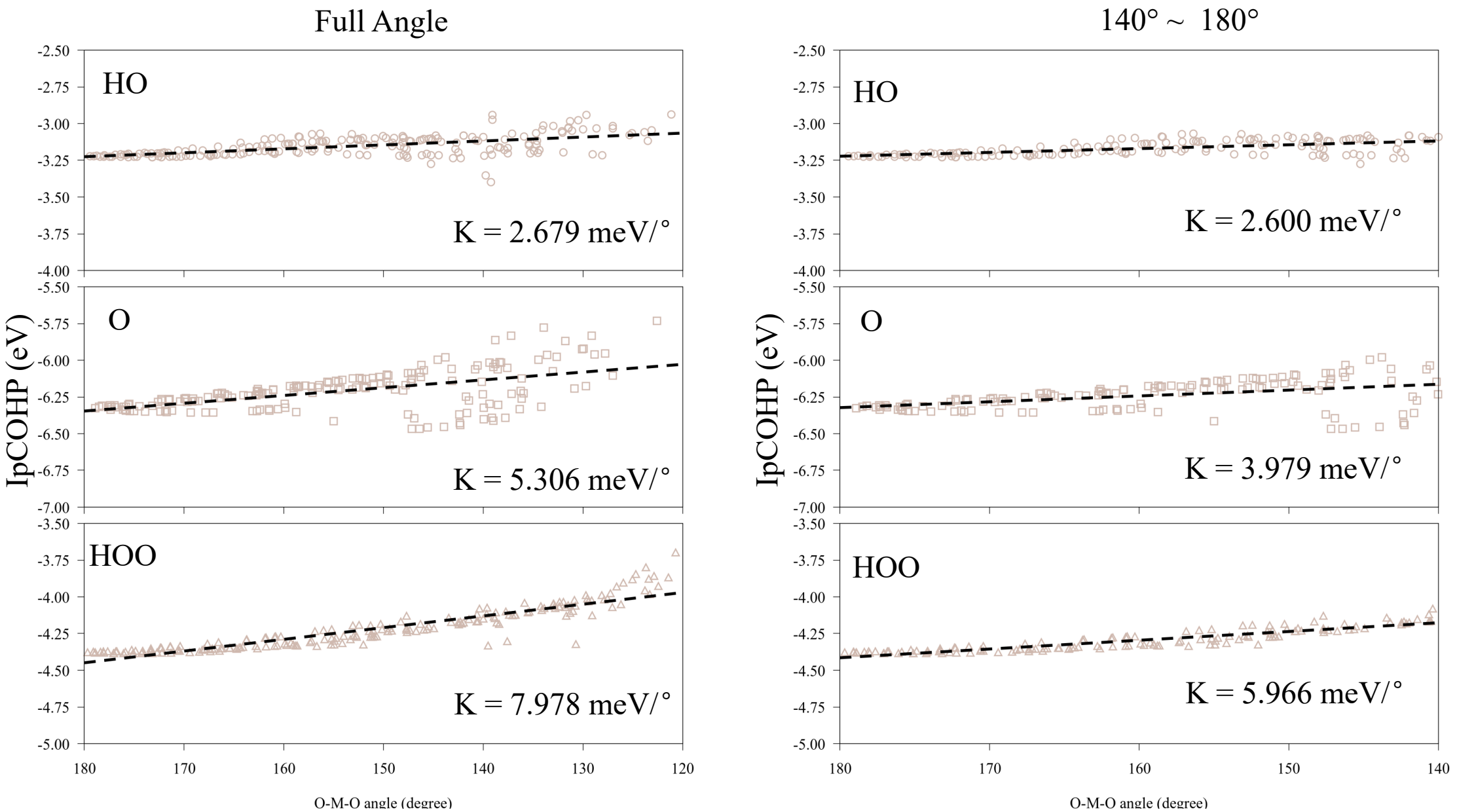
6.5 MnO_6

Slope: HO > HOO > > O



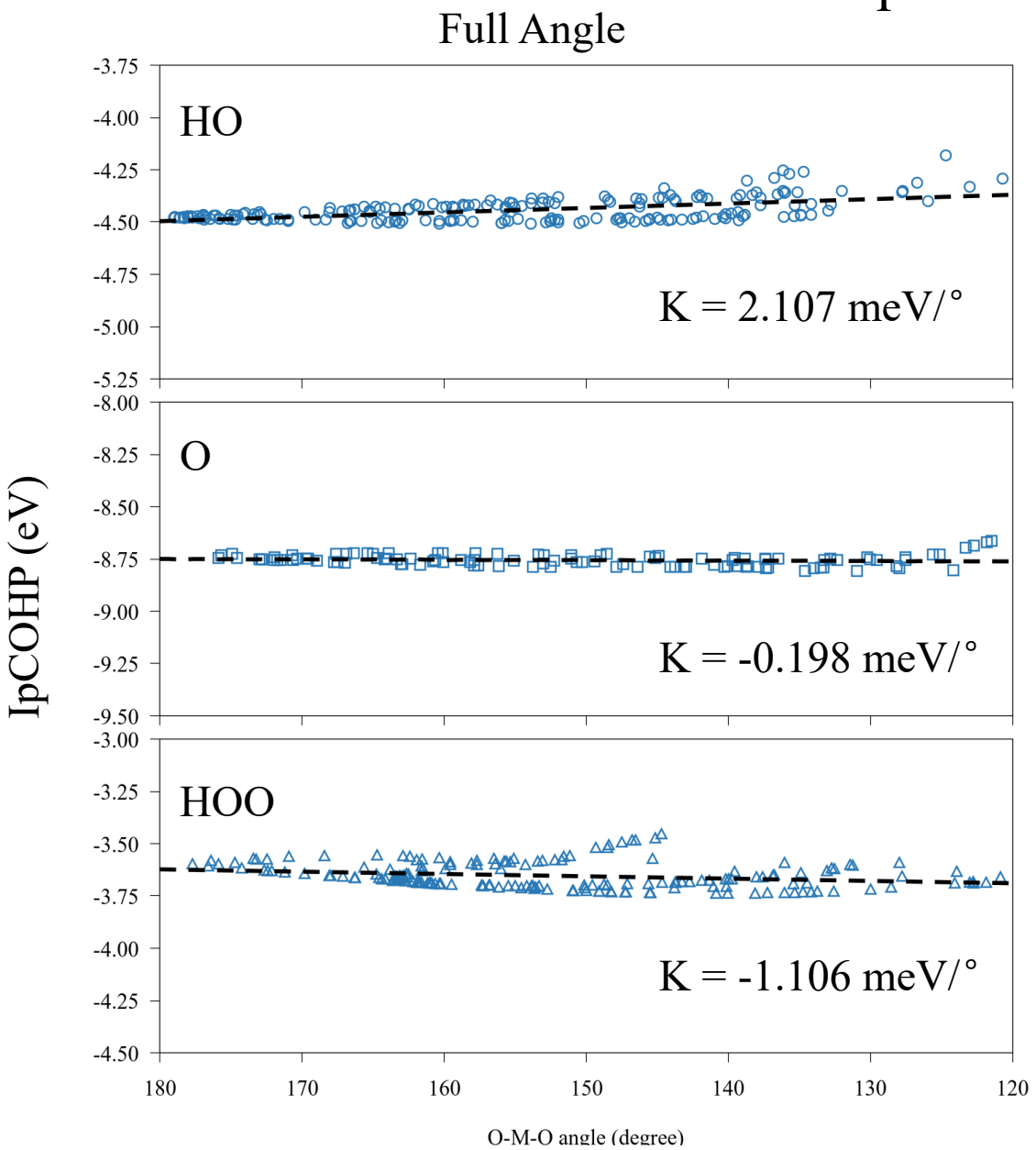
6.6 RuO_6

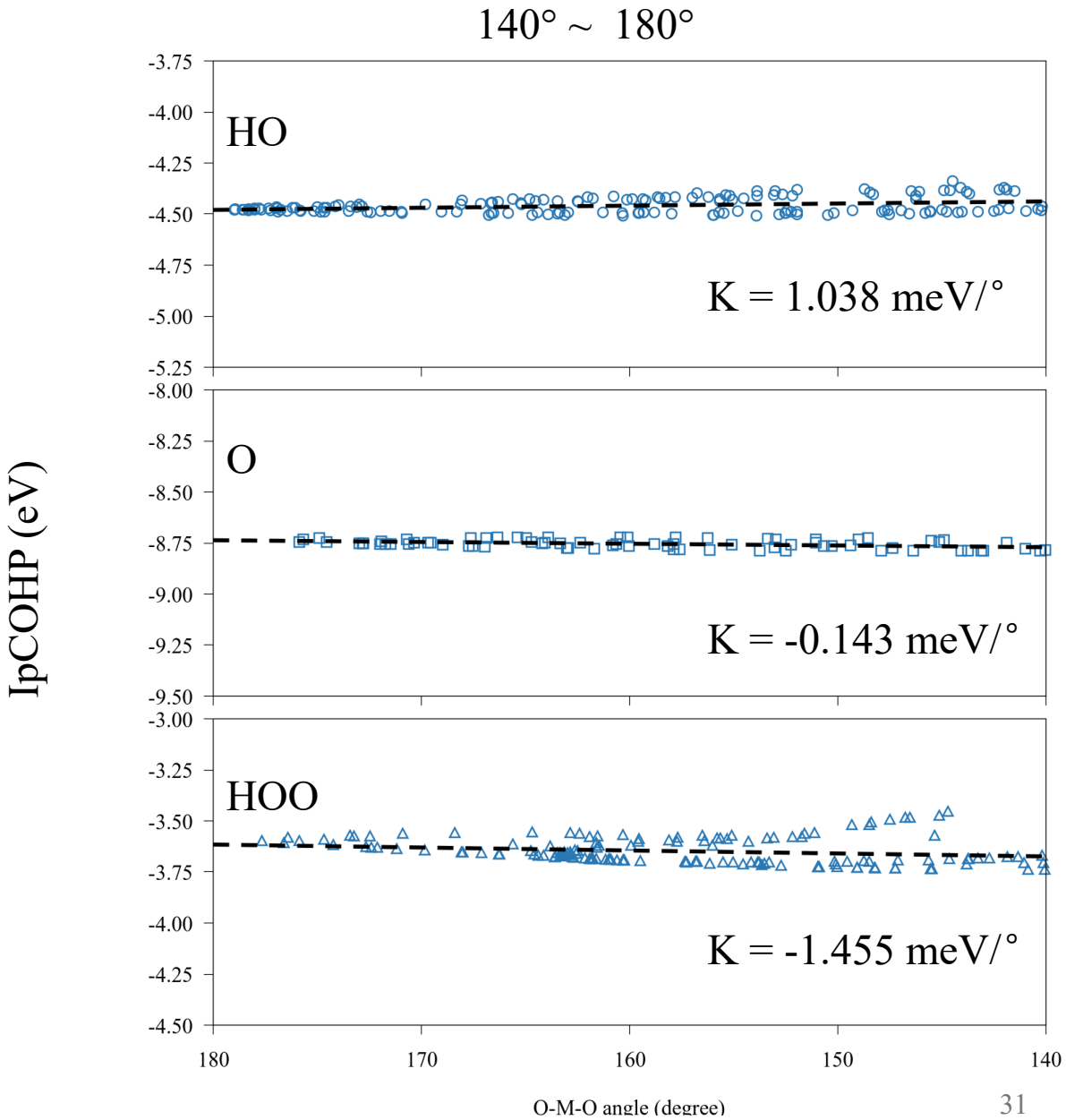
Slope: HOO > O > HO



6.7 TiO_6

Slope: HO > O > HOO





Expected trends

- HOO stronger than HO → $\Delta G_2 + \Delta G_3$ increases above 3.2 eV.
- HO stronger than HOO → $\Delta G_2 + \Delta G_3$ decreases below 3.2 eV.

Expected Slopes based on relative rates of change

- O changes more than HO and HOO → ΔG_3 decreases, ΔG_2 increases.
- O changes less than HO and HOO → ΔG_2 decreases, ΔG_3 increases.
- HO > HOO > O → ΔG_2 decreases, ΔG_1 increases.
- HOO > O > HO → ΔG_3 increases.
- HO > O > HOO → ΔG_1 increases.

6.8 Analysis

Slope of O–M–O in 120° ~ 180°

Element	HO (meV/°)	O (meV/°)	HOO (meV/°)	Slope Change
Ir	13.400	15.604	13.289	O > HO ≈ HOO
Ni	14.744	15.040	14.430	O >≈ HO ≈ HOO
Co	12.062	7.604	11.681	HO ≈ HOO > O
Fe	10.195	3.558	8.787	HO > HOO > O
Mn	6.743	0.143	3.778	HO > HOO > O
Ru	2.679	5.306	7.978	HOO > O > HO
Ti	2.107	-0.198	-1.106	HO > O > HOO

- Ir, Ni : O changes more than HO and HOO → ΔG_3 decreases, ΔG_2 increases.
- Co, Fe : O changes less than HO and HOO → ΔG_2 decreases, ΔG_3 increases.
- Mn : HO > HOO > O → ΔG_2 decreases, ΔG_1 increases.
- Ru : HOO > O > HO → ΔG_3 increases.
- Ti : HO > O > HOO → ΔG_1 increases.

7. Conclusion and Future Plan

Element	Oxide	PDS in Ref.	Angle change
Ir	IrO_2	ΔG_3 [1]	$\Delta G_3 \downarrow, \Delta G_2 \uparrow$.
Ni	NiOOH / NiO	ΔG_2/ ΔG_3 [2]	$\Delta G_3 \downarrow, \Delta G_2 \uparrow$.
Co	Co_3O_4 / CoOOH	ΔG_1 / ΔG_2/ ΔG_3 [3]	$\Delta G_3 \uparrow, \Delta G_2 \downarrow$.
Fe	FeOOH / Fe_2O_3	ΔG_2/ ΔG_3 [4]	$\Delta G_3 \uparrow, \Delta G_2 \downarrow$.
Mn	MnO_2	ΔG_2 [5]	$\Delta G_1 \uparrow, \Delta G_2 \downarrow$.
Ru	RuO_2	ΔG_3 [6]	$\Delta G_3 \uparrow$,
Ti	TiO_2	ΔG_1 [7]	$\Delta G_1 \uparrow$,

For MO_6 structures containing Ir, Ni, Co, Fe, and Mn, OER catalytic activity can be tuned by adjusting the O–M–O bond angles.

[1] Binninger et al., Energy Environ. Sci. 2022, 15 (6), 2519–2528.
[2] Wang et al., Nat. Commun. 2021, 12 (1), 6089.
[3] Wang et al., Nat. Commun. 2022, 13 (1), 6650.
[4] Song et al., ACS Cent. Sci. 2019, 5 (3), 558–568
[5] Zhang et al., J. Phys. Chem. C 2025, 129 (10), 5260–5267.
[6] Elkamash et al., ACS Electrochem. 2026, 2 (4), 1016–1026.
[7] Malik et al., Phys. Chem. Chem. Phys. 2023, 25 (3), 2203–2211.

7. Conclusion and Future Plan

1. Check if adsorption energy and ICOHP show a linear relationship for other metals.

   (Ni, Fe, Co and so on)

2. Look at how different electron orbitals contribute to bonding.

   (M d–O 2p hybridization)

3. Compare our results with experiments and other theoretical studies.

4. Check if this relationship also works for other OER mechanisms.

   (LOM, OPM, IMOC and PDAM)

ご清聴ありがとうございました